INTRODUCTION TO KNOT THEORY

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Abstract. These notes are based on an undergraduate course taught at the University of Chicago Center in Paris in Spring 2025.

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1. Introduction

1.1. **What is a knot?** I propose as a starting point a sort of slogan: *a knot is a tangled filament*. Untangling this definition requires us to expand or gloss the terms (tangled, filament) recursively into terms simple enough to test against our intuition. In place of the term 'tangled' we might substitute 'entwined', 'twisted', 'coiled', 'snarled', 'muddled' etc. Some of these terms suggest the result of performing *specific geometric operations* (twining, twisting, coiling) that implicitly take place in an ambient (three dimensional) space, while others suggest an idea of *topological complexity* which might take the form of an *obstruction* to an (as yet) undefined notion of *elementary* simplification. In place of the term 'filament' we might substitute 'fiber', 'thread' or 'strand' — what is apparently important here is the *one dimensionality* of a filament (especially when combined with the three dimensionality of its ambient embedding).

Our slogan reveals (or, anyway, suggests) two important aspects of knothood:

(1) it involves the ways that *one dimensional objects* can be embedded in *three dimensional space*; and
(2) it involves a *calculus* of *local topological operations* that create complexity or reduce it.

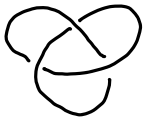
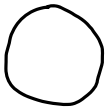
A different starting point, arising from the intuition that knots are geometric objects of some sort, would be to draw some pictures and then try to figure out what we have drawn:

Date: May 6, 2025.

 is this a knot?  what about this?  or this?

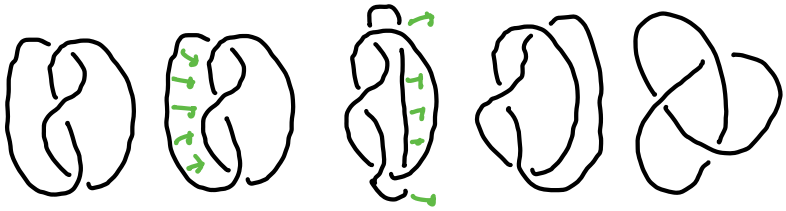
Of the three diagrams in this figure the one on the left certainly seems more 'snarled' or 'muddled' than the other two, which are progressively less so. We are likely to intuit some kind of topological distinction between the first figure and the third figure. On the other hand, our intuition easily fills in these discrete images with a continuous family of deformations, in which the rightmost 'end' (in the first figure) is pulled through a loop and then up and to the left around the strand emanating from the leftmost 'end'; this sense of continuous interpolation is at odds with our sense that the first and third picture are topologically distinct.

The 'trouble' here may be traced to the special role of the 'ends' of the strand; if they are allowed to freely move through space it is hard to see where any combinatorial or discrete notion of knottedness may reside. This trouble may be resolved by taking the two free ends of a strand and joining them up, so that our knot is topologically an embedded *circle* in space. Hence

 this is a knot  . . . but so is this!

These definitely look different, and indeed they are, in a sense that will take us a little while to make precise. The diagram on the left 'is' the right handed trefoil, while the diagram on the right 'is' the unknot. The word 'is' is in quotes because the diagram on the left is *not* a knot; it is a *picture of a knot*. This Magritte-like distinction goes in fact right to the heart of what knot theory is all about.

Consider the following sequence of diagrams:

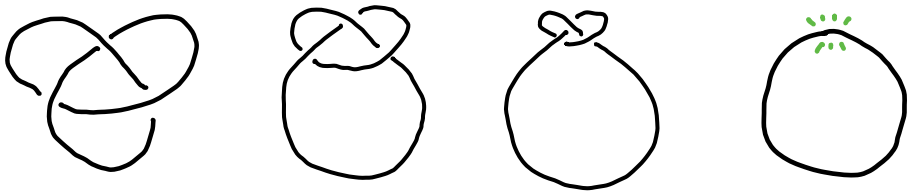


The diagrams at the extreme left and right certainly look different, at a glance. But the intermediate diagrams show how one may take the leftmost strand of the left diagram and pull it behind and to the right of the other strands in order to transform the first diagram into the last one.

This is *not* a problem to be remedied: the leftmost and rightmost diagrams are *both* pictures of the 'same' knot, 'the' right handed trefoil. Our geometric intuition tells us that the intermediate steps represent moves that may be accomplished by a continuous deformation of a knot in three dimensional space, even if this ends up producing a decisive change in the diagrammatic representation of the knot.

A knot then is an *equivalence class* of embedded circles in 3-dimensional space, where two circles are equivalent (i.e. represent the same knot) if they may be interpolated by a 'suitable' continuous family of embedded circles. The embedded circles themselves are *representatives* of the knot; our language in the sequel will not always make this distinction precise, but we should make sure we always understand it is there.

Why do I say 'suitable' continuous family? Why don't we define two embedded circles to be equivalent (i.e. to define the same knot) if they may be interpolated by *any* continuous family of embedded circles? Consider the following sequence of diagrams:

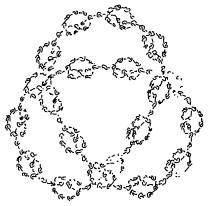


It is hard to dispute that this is a continuous family of embedded circles that interpolates between the right handed trefoil (on the left) and the unknot (on the right). All the 'knottedness' has been shrunk down to a point and disappeared. If transformations such as these are permitted, one suspects that any two knots will turn out to be equivalent, and knot theory will become trivial.

So: some kinds of continuous deformations are allowed, and some are not. To decide which are which we must now (unavoidably, I think) become slightly more formal and technical.

One loose end we should tie up first is the question of what exactly is the 3-dimensional space in which our knots are embedded. One natural choice is to work in $\mathbb{R}^3$, ordinary 3-dimensional Euclidean space. This will do for now, although at some point it will be more natural to work in the compact space S^3, which may be thought of as the set of unit vectors in $\mathbb{R}^4$ or more abstractly as a space obtained from $\mathbb{R}^3$ by adding a single compactifying point 'at infinity'.

Now that we have decided where our knots are embedded we need to be more careful about what exactly we mean by an embedding and what we mean by a continuous family. In order to avoid certain pathologies it makes sense to restrict attention to knots of finite combinatorial complexity (whatever that means). An example of the kind of pathology we do not allow is this (figure adapted from an original by Alberto Verjovsky):



This object is an embedding of a circle in $\mathbb{R}^3$. It may be obtained by an infinite recursive construction very similar to the construction of the (middle third) Cantor set. It is an

example of what is known as a *wild knot*. Wild knots and their theory are very interesting, but we do not study them here; we shall rule them out by fiat.

One way to do this is to restrict attention to *smooth* knots — i.e. knots with representatives that are the images of a smooth map $f : S^1 \to \mathbb{R}^3$ — i.e. the coordinate functions are continuous with continuous derivatives of all orders — which is both a topological embedding (i.e. injective) and an immersion (i.e. the vector f' is nowhere zero). And we may now say that two smooth knot representatives are *equivalent* if they are the images of smooth maps $f_0, f_1 : S^1 \to \mathbb{R}^3$ for which there is a smooth map $F : S^1 \times [0,1] \to \mathbb{R}^3$ for which the factors $f_t := F(\cdot, t) : S^1 \to \mathbb{R}^3$ are all themselves smooth knot representatives. This rules out the phenomenon observed in the figure above: although each knot in the family is the image of a smooth map $S^1 \to \mathbb{R}^3$, the second derivatives of these maps must necessarily blow up as the 'knotted' part shrinks to a point, and therefore the family of embedded circles cannot be parameterized by a smooth $S^1 \times [0,1] \to \mathbb{R}^3$.

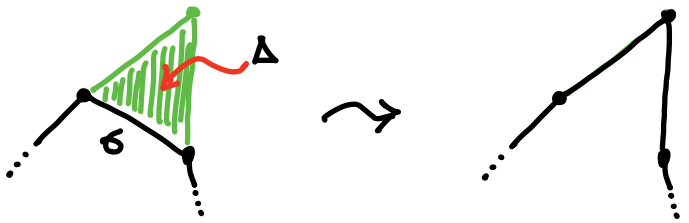
Example 1.1. The *unknot* is the equivalence class of smooth knots represented by some (any) round circle in $\mathbb{R}^3$ (exercise: these are all equivalent).

Example 1.2. For p, q coprime integers the (p,q)-torus knot is the knot represented by the map $f : S^1 \to \mathbb{R}^3$ with coordinates $f(\theta) = (r\cos(p\theta), r\sin(p\theta), -\sin(q\theta))$ for $r = 2+\cos(q\theta)$ and $\theta \in S^1 = \mathbb{R}/2\pi\mathbb{Z}$. The example $p = 2$, $q = -3$ gives the right-handed trefoil.

Another way to achieve a similar result is to restrict attention to *polygonal* knots — i.e. knots with representatives which are the union of a finite sequence of straight edges (the meaning of 'straight' here implicitly uses the linear structure of $\mathbb{R}^3$). And we may now say that two polygonal knot representatives are *equivalent* if one is obtained from the other by a finite sequence of combinatorial moves of two kinds:

(1) subdivision of an edge (or its inverse); or
(2) pushing an edge across a disjoint triangle (or its inverse).

The meaning of the first (subdivision) move should be clear. The second move means: we take an edge σ of a polygonal circle C which is simultaneously an edge of a triangle Δ that is disjoint from C except along $\sigma \subset \partial\Delta$. We may then replace C with the new polygonal circle C' defined by $C' := (C - \sigma) \cup (\partial\Delta - \sigma)$. I.e.:



If we need to refer to the second move by name we shall call it a 'Delta move' (but we shall rarely do this). Note that the Delta move increases the number of edges by one. Note too that subdivision is really just a special case of the Delta move for a degenerate triangle.

Which method shall we adopt? Formally we shall adopt the second method: a *knot* will be an *equivalence class* of *embedded polygonal circle* in $\mathbb{R}^3$, where two polygonal circles are

equivalent if one may be obtained from the other by a finite sequence of *subdivision* or *Delta moves* or their inverses.

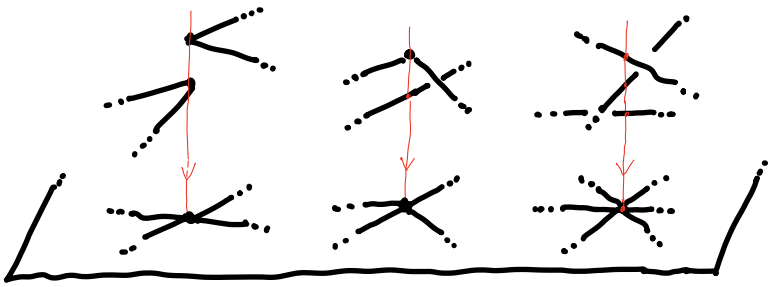
On the other hand, any polygonal knot may be approximated by a smooth knot, and any polygonal equivalence achieved by subdivision and Delta move may be approximated by a smooth interpolating family of smooth knots. Conversely, smooth knots may be approximated by polygonal knots, and if $F : S^1 \times [0,1] \to \mathbb{R}^3$ is a smooth family, we may subdivide $S^1 \times [0,1]$ into a fine polygonal mesh which corresponds to a sequence of subdivisions and Delta moves relating two polygonal approximations of $f_0(S^1)$ and $f_1(S^1)$. Thus we shall feel free (though we have not at all rigorously proved the claims of this paragraph!!) to work either with smooth or polygonal representatives of knots, and to use either smooth or polygonal equivalence relations between representatives to prove theorems about knots, whichever is most convenient.

1.2. **Knot diagrams and Reidemeister's Theorem.** At this point we need to confront a rather important issue: what exactly are these pictures we have been drawing up to now, and what exactly is their relationship to circles (smooth, polygonal or otherwise) in $\mathbb{R}^3$?

An embedded polygonal circle C in $\mathbb{R}^3$ may be projected orthogonally to a plane $\mathbb{R}^2$. Let $\pi : \mathbb{R}^3 \to \mathbb{R}^2$ denote the projection. We consider projections that satisfy some *nondegeneracy conditions*:

(1) distinct vertices of C map to distinct points of $\mathbb{R}^2$ (in particular, no edge of C is perpendicular to $\mathbb{R}^2$);
(2) if e is an edge and v is a vertex of C disjoint from e then $\pi(v)$ is disjoint from $\pi(e)$; and
(3) no point of C is in the image of 3 distinct points of C.

The three kinds of degeneracy are exhibited in this figure:



Exercise 1.3. Show that for any embedded polygonal circle C in $\mathbb{R}^3$ the set of orthogonal projections satisfying the nondegeneracy conditions is open and nonempty (in what?)

The nondegeneracy conditions imply that the projected circle $\pi(C)$ is in general position: it crosses itself transversely at only finitely many points, and there are no triple points. Since each self-intersection point of $\pi(C)$ has exactly two preimages on distinct edges of C we record the information of which edge is 'above' the other (this is indicated pictorially by omitting a small part of the 'lower' edge). This data — an immersed polygonal circle $\pi(C)$ together with crossing data — is a (polygonal) *knot diagram*.

We shall be a little bit careless about the polygonal structure of $\pi(C)$ and admit the possibility that a sequence of distinct consecutive edges of C may project to parallel segments in $\pi(C)$ that appear (in the diagram) as a single polygonal edge. With this caveat the following is true:

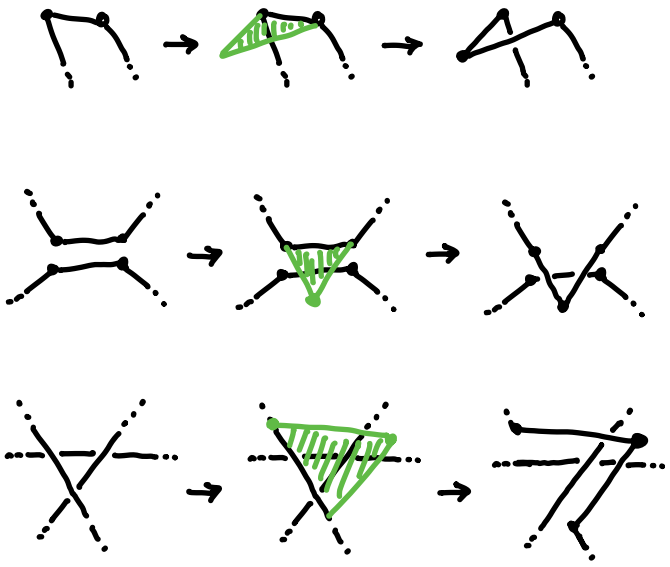
Exercise 1.4. Show that a polygonal knot diagram determines a unique equivalence class of polygonal knot.

Now, let's suppose that C and C' are polygonal knots related by a finite sequence of subdivision and Delta moves or their inverses, and let's further suppose that $\pi(C)$ and $\pi(C')$ are valid knot diagrams. What is the relationship between $\pi(C)$ and $\pi(C')$? The answer is given by the following theorem of Reidemeister:

Theorem 1.5 (Reidemeister)**.** *The diagrams $\pi(C)$ and $\pi(C')$ determine equivalent classes of polygonal knots if and only if $\pi(C')$ may be obtained from $\pi(C)$ by a finite sequence of intermediate diagrams which either do not change the topology of the diagram (we say they are achieved by* planar isotopy*) or one of the following three moves or their inverses:*

(1) the first Reidemeister move creates one new crossing and a new 'monogon' complementary region;
(2) the second Reidemeister move creates two new crossings and a new 'bigon' complementary region; and
(3) the third Reidemeister move 'pushes' an overcrossing strand over a crossing of two lower strands;

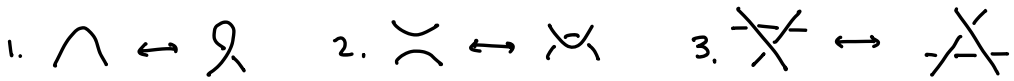
The following three figures (plus obvious variations obtained by reflection or rotation) illustrate these moves:



Exercise 1.6. Prove Reidemeister's theorem! The figures above will give you a strong hint how to do this. Suppose we want to accomplish a Delta move. A priori the projection of the triangle Δ to $\mathbb{R}^2$ may intersect many different segments and crossings of the projection.

But if we subdivide Δ sufficiently finely, we may arrange that the projection of each triangle in the subdivision (and hence the change to the diagram associated to a Delta move) has bounded combinatorial complexity falling into finitely many cases that may be enumerated.

One can also define knot diagrams and Reidemeister moves in the smooth world. There are the same three combinatorial moves (plus planar isotopy) and in practice we tend to draw smooth knot diagrams because they look nicer. The 'smooth' version of the Reidemeister moves is illustrated in the following figure:



Reidemeister's Theorem 'reduces' knot theory to a calculus of immersed circles in the plane with crossing data together with a finite set of combinatorial transformations between such circles.

Exercise 1.7. What are the analog of the nondegeneracy conditions for smooth embedded circles in $\mathbb{R}^3$?

1.3. **Knot invariants.** We now have a workable definition: a knot is an equivalence class of (polygonal or smooth) embedded circle in $\mathbb{R}^3$ up to the relation of (polygonal or smooth) equivalence. Nondegenerate projection to a plane sets up a bijection between knots and equivalence classes of knot diagrams up to the relation generated by planar isotopy and Reidemeister moves.

Now that we know what knots are, and have at least two rather distinct ways of talking about them, we may start to do knot theory. The most basic question one might ask is simply: what knots are there? It is not clear at the outset what counts as an answer. The class of all knots forms a (nonempty!) set, and we can ask how many elements there are in this set. There are evidently only countably many equivalence classes of (polygonal) knot diagrams, so there are countably many knots; but are there finitely many or infinitely many? Are there more than one? Can we enumerate them? How would we get started addressing such questions?

Since the definition of knots depends on a rather opaque equivalence relation we might want to focus our initial attention there, and ask: given two knot representatives, how can we tell whether they represent the same knot? Or: how can we tell whether they represent different knots?

Let's let $\mathcal{K}$ denote the set of all knots, and $\mathcal{D}$ the set of all knot diagrams. Let $\sim$ be the equivalence relation on $\mathcal{D}$ generated by planar isotopy and Reidemeister moves, so that there is a bijection $\mathcal{K} \to \mathcal{D}/\sim$. A *knot invariant* is a function $x : \mathcal{D} \to X$ which is constant on the equivalence classes of $\sim$. It thereby induces a function which by abuse of notation we denote $x : \mathcal{K} \to X$, and if we have knots K_1, K_2 with $x(K_1) \neq x(K_2)$ we can deduce that $K_1 \neq K_2$.

Let's consider some examples.

Example 1.8 (Knot type). The simplest invariant is just the map $\mathcal{D} \to \mathcal{K}$ that takes a diagram to its underlying knot type. This map inverts the bijection $\mathcal{K} \to \mathcal{D}/\sim$, a fact

which is equivalent to Reidemeister's Theorem. While in principle this invariant is the most powerful (in the sense that any knot invariant may be obtained as a factor of this one) it is essentially useless as a tool for calculation, since it just hands our knot back to us with no more information.

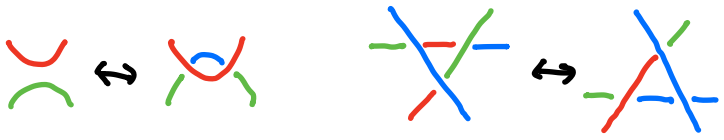
Example 1.9 (Crossing number). Suppose we have a numerical function f on $\mathcal{D}$ which takes values in a well-ordered set. Then we may define a knot invariant by $D \to \min_{D' \sim D} f(D)$. For example, there is an invariant $c : \mathcal{D} \to \mathbb{N}$ which counts the number of crossings in a diagram. The *crossing number* of a knot K (denoted $c(K)$) is then the *minimum number of crossings* in any diagram representing K. At first glance an invariant of this kind might seem useless, since there is no apparent way to calculate it. One thing that is easy to compute is an upper bound: $c(K) \leq c(D)$ for any diagram D representing K. In particular, $c(K) = 0$ for the unknot.

Example 1.10 (3-colorings). A *segment* of a diagram D is an arc of the diagram running between two consecutive undercrossings. Other arcs of the diagram may cross under a segment, but no arc may cross over it (by definition). If you draw a diagram D, the segments are the continuous strokes that may be made without lifting the pen (or chalk as the case may be). Note that if $c(D) > 0$ (i.e. there is at least one crossing) then D has $c(D)$ segments.

Let's fix three distinct colors (for the sake of argument, let the colors be red, green and blue). A *3-coloring* of a diagram D is an assignment of one of the colors to each segment so that at each crossing, the three segments that meet at the crossing either all have the same color, or they all have different colors. Let $\tau_3(D)$ denote the number of 3-colorings of a diagram D.

Proposition 1.11. *$\tau_3(D)$ is a knot invariant. In other words, if $D \sim D'$ then $\tau_3(D) = \tau_3(D')$.*

Proof. Changing a diagram by planar isotopy does not change the number of 3-colorings. Thus to prove the proposition it suffices to show that $\tau_3(D) = \tau_3(D')$ whenever D and D' differ by a single Reidemeister move. This may be verified by a finite calculation; two diagrams that differ only in the region associated to a move may be colored the same way outside this region, and inside this region a coloring of one diagram determines a coloring of the other. For example:



□

Let's compute τ_3 for some simple knots.

Some trial and error shows that the unknot and the figure 8 knot admit only monochromatic colorings, so $\tau_3 = 3$ for both of those. The right handed trefoil however admits a non-monochromatic coloring, and in fact $\tau_3 = 9$. In particular: *the right-handed trefoil is not equivalent to the unknot*!



What about the unknot and the figure 8 knot? Are they equivalent? The question cannot be decided by τ_3 alone; the invariant is not powerful enough to distinguish them. Of course, at this point as far as we know, this might be because the knots are actually equivalent; this turns out not to be the case, but we must work harder to show it.

Exercise 1.12. Show that $\tau_3(K)$ is always a power of 3.

In general, what does τ_3 tell us about a knot? We are not yet in a position to answer this question in full generality. Nevertheless we can relate τ_3 in a useful way to crossing number $c(D)$. If $c(D) = 0$ or 1 then there is only one segment, and evidently $\tau_3(D) = 3$. Suppose $c(D) > 0$ so that $c(D)$ is equal to the number of segments of D. Each segment may be colored in at most 3 ways, so $\tau_3(D) \leq 3^{c(D)}$. In fact it is not hard to see that if $c(D) \geq 2$ then $\tau_3(D) \leq 3^{c(D)-1}$. For, in this case there is at least one crossing at which at least two different segments meet. Either exactly two segments meet there (in which case both segments must necessarily have the same color) or three segments meet there (in which case the color of two of them uniquely determines the third). In either case the 3-coloring condition imposes at least one 'constraint' on the choice of colors, so that the number of 3-colorings is at most $3^{c(D)-1}$. Since this inequality holds for *every* diagram, and since τ_3 is a knot invariant, it follows that $c(K) \geq 1 + \log_3 \tau_3(K)$ providing $\tau_3(K) > 3$.

For the unknot this tells us nothing new, since we already know $c(K) = 0$ in this case; and for the figure 8 knot it tells us nothing at all. But for the right handed trefoil it gives the inequality $c(K) \geq 3$ which, together with the inequality $c(K) \leq 3$ (which may be deduced by exhibiting the standard diagram) shows that in fact $c(K) = 3$.

Determining the crossing number of a knot in general is a difficult problem. The extent to which we can make progress on this problem in the sequel will give us a rough estimate of the power of our methods.

The crossing number of the unknot is 0. Are there any other knots with crossing number 0? It seems unlikely. But to prove it we must develop some machinery, and this is the purpose of the next section.

2. Surfaces

2.1. The Jordan Curve Theorem.

In fact there is only one knot with crossing number zero — the unknot. This will follow from two facts. The first is a topological characterization of the unknot.

Proposition 2.1. *An embedded circle in $\mathbb{R}^3$ is the unknot if and only if it bounds an embedded disk.*

Proof. The unknot is represented by a round circle C which certainly bounds an embedded disk D. If C' is any other representative then C and C' are related by smooth equivalence; we claim that this implies that C' also bounds an embedded disk D'. Imagine that the

space $\mathbb{R}^3 - C$ outside C is filled with a malleable rubber, which may stretch indefinitely but does not tear or rip. As we move C around via the track of the smooth equivalence, the space outside compresses to avoid the space that C moves into, and stretches to fill the void it leaves. The disk D is also carried along by this process, and at the end of the process it gives a new embedded disk D' in $\mathbb{R}^3 - C'$ as claimed. This proves one direction of the proposition: every representative of the unknot bounds an embedded disk.

To prove the other direction, suppose C is a smooth circle bounding a smooth embedded disk D. To say that D 'is' a disk is to say that there is a smooth parameterization $f : D^2 \to D$ where D^2 is the 'standard' unit disk in $\mathbb{R}^2$, and where the Jacobian df of f is injective on the tangent space at every point of D^2 (this just means that D looks like a flat plane to first order near every point). In particular, if $\alpha \subset D^2$ is a very small round circle centered at 0 then $f(\alpha)$ is close (to first order) to a very small ellipse contained in a plane in $\mathbb{R}^3$. Such an ellipse is evidently equivalent to the unknot (by a family of affine transformations). On the other hand, α and ∂D^2 may be interpolated by a 1-parameter family of concentric round circles in D^2, and the images of these circles under f is a smooth equivalence between C and $f(\alpha)$. Hence C represents the unknot, as claimed. □

Exercise 2.2. Give a careful proof that a smooth equivalence from C to C' in $\mathbb{R}^3$ extends to a diffeomorphism (i.e. a smooth, smoothly invertible map) from $\mathbb{R}^3$ to itself taking C to C'. In fancy language, what you are doing is extending an *ambient isotopy of a subspace* to a *diffeomorphism of the ambient space*.

The second fact is rather more substantial

Theorem 2.3 (Jordan Curve Theorem)**.** *Every embedded circle in $\mathbb{R}^2$ bounds an embedded disk in $\mathbb{R}^2$.*

Let us first see that these facts imply the desired result.

Proposition 2.4. *Any knot with crossing number 0 is the unknot.*

Proof. Let K be a knot which has a diagram D with crossing number 0. This means there is a representative C in $\mathbb{R}^3$ with an orthogonal projection $\pi : C \to \mathbb{R}^2$ which is an embedding onto its image. But then $\pi(C)$ itself is an embedded circle, and C is (smoothly) equivalent to $\pi(C)$ by a family of circles obtained by pushing each point of C at constant speed to its image in $\pi(C)$. Now by the Jordan Curve Theorem 2.3 the circle $\pi(C)$ bounds an embedded disk in $\mathbb{R}^2$ and hence also in $\mathbb{R}^3$; thus by Proposition 2.1 the circle $\pi(C)$ is the unknot and therefore so is K. □

We shall now give the proof of the Jordan Curve Theorem 2.3.

Proof. We give a proof for polygonal circles, which are rather convenient for inductive arguments. The proof has two parts: first we show that a polygonal circle C in $\mathbb{R}^2$ induces a decomposition of $\mathbb{R}^2 - C$ into two pieces, one compact (the 'inside') and one noncompact (the 'outside'); and second we show that the compact piece is a disk.

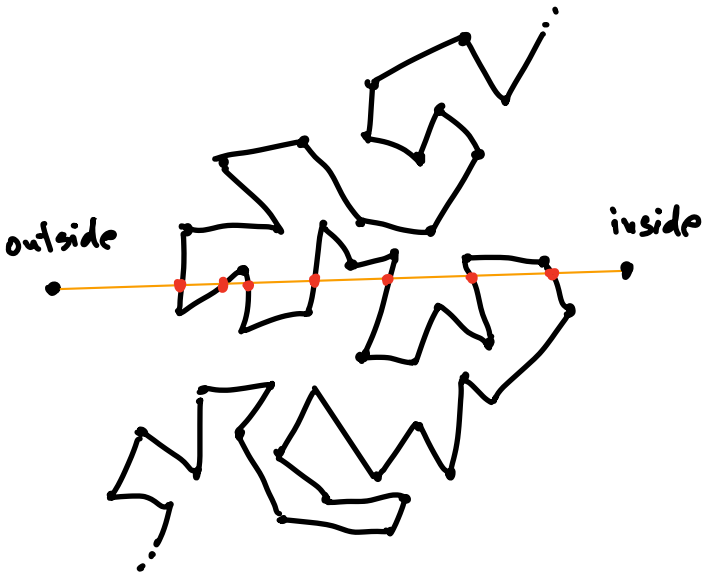
A tubular neighborhood $N(C)$ of C is an annulus; the circle C certainly divides $N(C)$ into two (thinner) annular components. If X is the result of cutting $\mathbb{R}^2$ open along the curve C then X is itself a (not necessarily connected) surface, with two boundary components which correspond to the two sides of C in $N(C)$. Each component of X must contain

at least one boundary component. This is because any point in $\mathbb{R}^2$ may be joined to C by a path, and some subset of this path joins any given point of X to a boundary point. It follows that $\mathbb{R}^2 - C$ has at most two pieces; we now show that it has at least two by exhibiting a nonconstant function ϕ from $\mathbb{R}^2 - C$ to $\{0, 1\}$ which is constant on each component.

Since C is compact, it is contained inside a ball of finite radius centered at the origin; without loss of generality, suppose it is contained inside the closed ball of radius $1/2$. Any two points outside this ball may be connected by a path outside this ball. Thus X has a unique noncompact component, which we will henceforth call the *outside*.

Let ℓ be a nearly horizontal line segment which is not parallel to any edge of C, and for any point $p \in \mathbb{R}^2 - C$ let ℓ_p be the unique segment parallel to ℓ joining p to some point p' with x-coordinate -1. By hypothesis $\ell_p \cap C$ is finite. At each point $q \in \ell_p \cap C$ we compute an *index* $i(q)$ which is either 0 or 1 as follows. Since (by hypothesis) q is necessarily an isolated point of $\ell_p \cap C$ the two 'sides' of C near q may either be above ℓ_p or below it. If they are both on the same side the index $i(q)$ is 0; otherwise it is 1 (notice that $i(q) = 0$ is only possible if q is a vertex of C). Now: define $\phi(q)$ to be equal to the sum $\sum_{q \in \ell_p \cap C} i(q)$ mod 2.

This figure shows an example for which p (on the right) is joined to p' (on the left) by a segment ℓ_p which intersects C at seven points, all with index 1. Hence $\phi(p) = 1$.



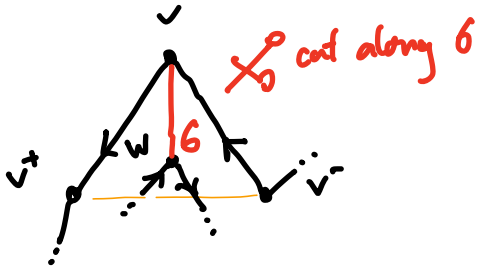
We claim ϕ is locally constant. Evidently it does not change if we slide p a little bit along the segment ℓ_p in either direction, since $\ell_p \cap C$ is constant. If we slide p slightly up or down, each point q of intersection of $\ell_p \cap C$ of index 1 slides up or down to a new point of intersection of index 1. If q is a point of intersection of $\ell_p \cap C$ of index 0, with the two sides of C near q on the positive side of ℓ_p (say), then sliding p up splits q open into two points of intersection each of index 1, whereas sliding p down eliminates the intersection q altogether. In either case the total value of ϕ does not change.

The function ϕ takes the value 0 on any point p with x coordinate -1, since in this case $p = p'$ and ℓ_p is a degenerate segments that does not intersect C at all. Thus $\phi = 0$ on

the entire outside component. On the other hand, it is easy to produce some point p with $\phi(p) = 1$ (just construct any segment parallel to ℓ that intersects C once in a point with index 1, and observe that the value of ϕ is different at the two endpoints). The set of all points in $\mathbb{R}^2 - C$ with $\phi = 1$ is therefore nonempty, and since none of these points is in the outside component, there must be at least two components, and therefore exactly two. This completes the first part of the proof.

Now let's prove that the compact component P (the 'inside') is topologically a disk. We prove this by induction on the number of vertices (this is a place where it is especially convenient to work with polygons rather than smooth circles). Any polygonal circle must have at least 3 vertices; if it has exactly 3 it is the boundary of a triangle, so that P is certainly a disk in this case.

Now suppose C has at least 4 vertices. We orient the circle C 'positively', so that as we move around C in the positive direction the inside (i.e. P) is always to the left. Pick a vertex $v \in C$ at which P has an acute angle and let v^+ and v^- be the two vertices adjacent to v so that the three vertices v^-, v, v^+ span a triangle Δ. If Δ is contained in P we may push P across Δ by a Delta move, reducing the number of vertices. Otherwise there must be some vertex w of C inside Δ (possibly on the edge joining v^+ to v^-) and therefore there must be some 'highest' vertex w in Δ so that the segment σ joining w to v is contained in the interior of P.



But now we may cut P along σ; this produces two polygonal circles C_1 and C_2, each with fewer vertices (to see this, observe that the orientation on C near w must be as indicated in the figure, since σ is 'to the left' of C at either end) and each of these must bound a compact region P_1 and P_2 respectively.

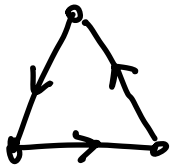
Evidently $P = P_1 \cup_\sigma P_2$. Since each of the C_i has fewer vertices than C, the regions P_i are both disks. Since P is obtained by gluing these disks along segments in their boundary, it is a disk too. The proof follows. □

2.2. **The Classification of Surfaces.** A *surface* is a space that looks locally 2-dimensional; i.e. every point has a little open neighborhood which is topologically equivalent to the open unit disk in the plane. One usually throws in some technical terms like 'countable basis' and 'Hausdorff' (if you know what those words mean); but we will soon narrow our focus so that these conditions hold automatically.

We shall be interested in surfaces that are *compact* and *triangulated*. In other words, they are obtained from a finite collection of triangles by gluing edges together in pairs. We shall eventually also allow surfaces with boundary (obtained by leaving some of the edges

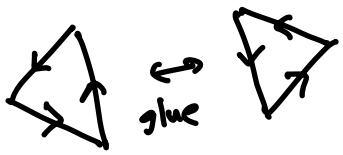
free and unglued) but for now let's focus on *closed* surfaces: those that are compact and without boundary.

We would also like to focus on *oriented* surfaces. What does this mean? We will take it for granted that we understand what it means to orient a triangle, and that a triangle may be oriented in two distinct ways, e.g. by choosing which of the two 'sides' of it is the 'top' and which is the 'bottom'. An oriented triangle induces orientations on the boundary edges, so that for each positively oriented edge on the boundary, the interior of the triangle lies to the *left* of the edge. In other words, the oriented boundary of the triangle goes anticlockwise around the interior.



Exercise 2.5. Make sense of what it means to say that 'the interior of the triangle lies to the left of an oriented boundary edge'. More precisely: explain how making sense of the word 'left' here is tantamount to *choosing* an orientation on the triangle.

When we glue up oriented triangles Δ_1 and Δ_2 along edges $e_i \subset \partial\Delta_i$ by some identification $f : e_1 \to e_2$, in order to get an oriented surface the identification f must *reverse* the orientations of the edges. This is because in the glued up surface S the triangle Δ_1 will be to the left of the glued up edge in one orientation, while Δ_2 will be to the left in the other orientation.



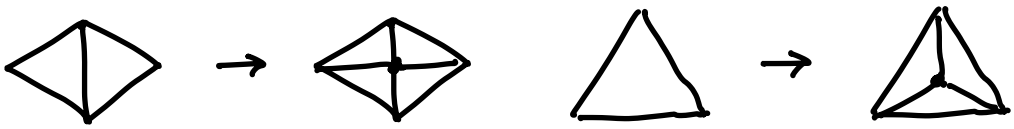
After performing gluings along all pairs of edges, the resulting surface may be thought of as a finite combinatorial object, which is specified by the finite amount of data consisting of the list of triangles, and the choice of pairing of edges. One might say the surface is a special kind of *two-dimensional simplicial complex*, for which every point is in the interior of a unique triangle, edge or vertex.

Definition 2.6 (Euler characteristic)**.** Let P be a two-dimensional simplicial complex. The *Euler characteristic* of P, denoted $\chi(P)$, is the alternating sum

$$\chi(P) := \#\text{triangles} - \#\text{edges} + \#\text{vertices}$$

A triangulation of a surface may be *refined* by a finite sequence of subdivisions, at every step subdividing either an edge or a triangle.

It is clear that the Euler characteristic does not change under refinement. Furthermore, it turns out that any two triangulations of a compact surface have a common refinement. To see this let S be the surface, and τ_1, τ_2 the triangulations. Wiggle the edges of τ_2 so that



they are transverse to the edges of τ_1, so that the union of the edges cuts up the surface into finitely many regions bounded by polygonal circles. Each one of these regions is contained in a triangle of either of the original triangulations, and is therefore topologically a disk (by the Jordan Curve Theorem 2.3) and may therefore be subdivided into triangles by coning the boundary to some point in the interior, producing a new triangulation τ whose edges 'contain' the edges of both τ_1 and τ_2.

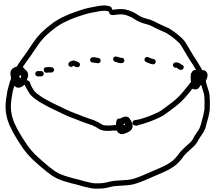
Exercise 2.7. Show that τ as above admits a common refinement with either of τ_1 or τ_2.

We may conclude that the Euler characteristic is a topological invariant of a compact surface, and henceforth we will write $\chi(S)$, to emphasize that it is independent of the choice of a triangulation.

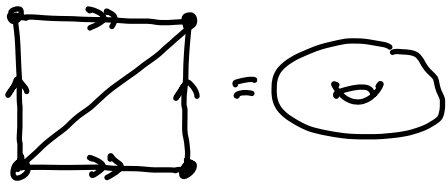
Let us now state our first version of the Classification of Surfaces:

Theorem 2.8 (Classification of Closed Oriented Surfaces)**.** *Let S be a closed oriented connected surface. Then $\chi(S) = 2 - 2g$ for some non-negative integer g called the* genus *of S. This may take any non-negative value, and any two closed oriented connected surfaces with the same genus are topologically equivalent.*

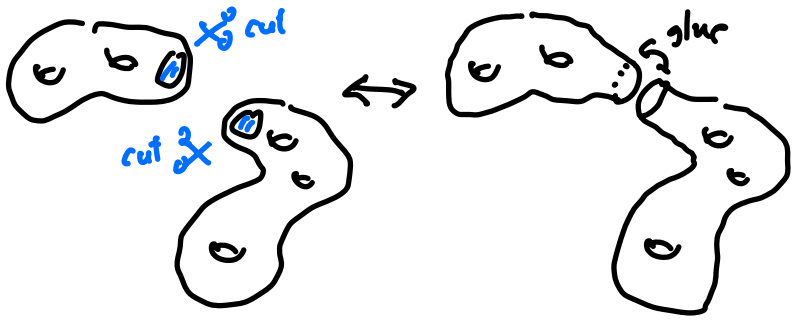
Example 2.9 (The 2-sphere). The 2-sphere S^2 may be triangulated by subdividing the equator into three segments, and then thinking of both the northern and the southern hemispheres as triangles for which these three segments are the boundary edges. Thus $\chi(S^2) = 2 - 3 + 3 = 2$.



Example 2.10 (The torus). The torus T may be obtained from a square by identifying opposite sides by translation. As such it admits an obvious triangulation (inherited from a triangulation of a square) with two triangles, three edges and one vertex. Thus $\chi(T) = 2 - 3 + 1 = 0$.



Example 2.11 (Connect sum). Suppose S_1 and S_2 are two closed, oriented, connected surfaces. We may build a new surface called the *connect sum* of S_1 and S_2, and denoted $S_1 \# S_2$, by removing the interior of a closed disk from each surface and gluing the resulting boundary circles (oriented as the boundaries of oriented surfaces) by an orientation-reversing identification.



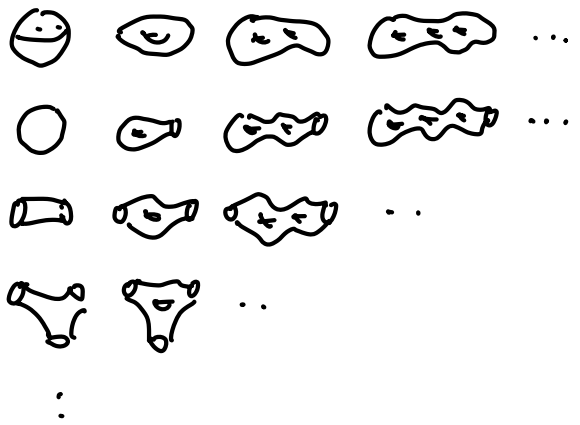
As triangulated surfaces this may be accomplished by removing the interiors of a single triangle from each of S_1 and S_2 and gluing the exposed edges. Removing two triangles reduces χ by 2; identifying two boundary circles by gluing identifies three pairs of edges and three pairs of vertices, and therefore it does not change χ. Thus $\chi(S_1 \# S_2) = \chi(S_1) + \chi(S_2) - 2$. In particular, by taking the connect sum of a torus with itself g times we may obtain a closed oriented connected surface with $\chi = 2 - 2g$.

Exercise 2.12. Show that the result of connect sum is well-defined (i.e. the result does not depend topologically on what the disks are that are removed or how the gluing is done), and that $\#$ defines an associative and commutative operation on topological equivalence classes of closed oriented connected surfaces, with S^2 as an identity element.

The main point of Exercise 2.12 is to show that there is only one way (topologically) to remove the interior of a closed disk from a connected surface. We may use this observation to deduce the classification of compact oriented surfaces from the classification of closed ones:

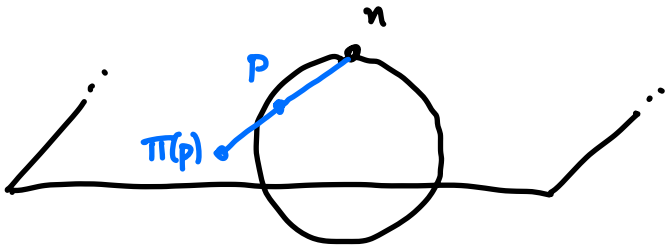
Theorem 2.13 (Classification of Compact Oriented Surfaces)**.** *Let S be a compact oriented connected surface with b boundary components. Then $\chi(S) = 2 - 2g - b$ for some non-negative integer g (the genus), and S is obtained from a closed oriented surface S' of genus g by removing the interiors of b disjoint closed disks. The numbers g and b may take any non-negative values, and any two compact oriented connected surfaces with the same genus and number of boundary components are topologically equivalent.*

Proof. If S is a compact surface with b boundary components then we may obtain a *closed* surface S' by attaching a disk to each of these b boundary components. If S is oriented, so is S'; and if S is connected, so is S'. By Theorem 2.8 the surface S' is classified by its genus g, and therefore S is classified by the genus g and the number b of boundary components. If S' is triangulated, we may obtain S by removing the interiors of b disjoint triangles; thus $\chi(S) = \chi(S') - b = 2 - 2g - b$. □



Theorem 2.8 is a powerful theorem. Before we prove it, we shall give a sense of its power by using it to give a quick new proof of the Jordan Curve Theorem.

Proof. Let C be a circle in $\mathbb{R}^2$. Now, $\mathbb{R}^2$ is not compact, but we may 'compactify' it by adding a point 'at infinity' to obtain S^2. The inverse of this operation is stereographic projection: if we think of S^2 as the round sphere in $\mathbb{R}^3$ and $n \in S^2$ as the north pole, for every other point $p \in S^2$ there is a unique straight line ℓ_p containing n and p, and this line intersects the x–y plane at a unique point $\pi(p)$. The map $\pi : S^2 - n \to \mathbb{R}^2$ is a diffeomorphism.



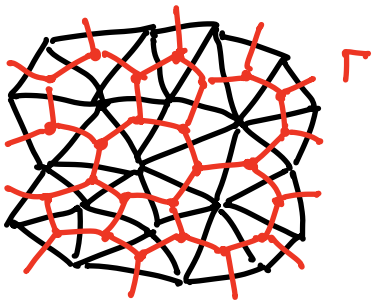
Thus the Jordan Curve Theorem will follow from a stronger claim: if C is a circle in S^2, it divides it into two regions, both of which are disks. To prove this, cut open S^2 along the circle C to obtain a compact oriented surface Σ. Gluing two boundary circles does not change χ so $\chi(\Sigma) = 2$. The surface Σ has two boundary components, and as we argued earlier, each component of Σ must contain at least one boundary component; hence Σ consists of at most two components. Let Σ' be obtained from Σ by attaching a disk to each boundary component. Then Σ' is closed and oriented, and $\chi(\Sigma') = 4$. Since Σ' has at most two components, and by Theorem 2.8 each of these has Euler characteristic at most 2, it must have *exactly* two components each with $\chi = 2$ and therefore each a sphere. Thus Σ has two components, each with one boundary component and $\chi = 1$, which are therefore both disks. □

Let us now give the proof of Theorem 2.8.

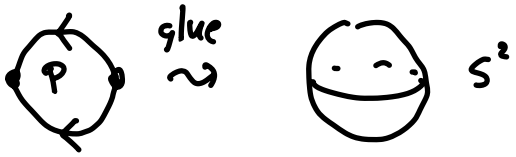
Proof. Let Σ be a closed, oriented, connected surface. We assume Σ is triangulated; i.e. it is obtained by gluing together the edges of finitely many triangles. The first step is to

show that Σ may be obtained from a single (planar) *polygon* with an even number of edges by gluing these edges in pairs.

In the triangulated surface Σ we obtain a graph Γ which is the *dual* of the 1-skeleton of the triangulation τ of Σ. This means that Γ has one vertex in each triangle of τ, and one edge that crosses each edge of τ. Since Σ is connected so is Γ. We may therefore choose a maximal tree $T \subset \Gamma$ that contains all the vertices. The edges of τ 'dual' to the edges of T correspond to a subset of the pairs edges of our original collection of triangles; if we glue up these pairs of edges (and only these) we obtain a polygon P (exercise: prove this!)



Since Σ is oriented, so is P and therefore so are the edges of ∂P, and these edges are glued together in pairs by orientation reversing identifications. The number of edges of P is positive and even. If it equals 2 there is only one possibility and the result of gluing these edges together is a sphere.



Otherwise P can be cut up into some positive number n of triangles by adding $n-1$ new interior edges, and hence

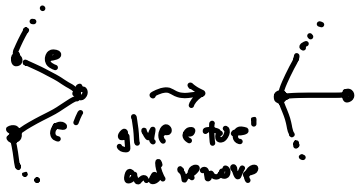
$$\chi(\Sigma) = 1 - \frac{1}{2}\#\text{edges of } P + \#\text{vertices after gluing}$$

Lemma 2.14. *We may modify P so that either P has only two edges (and therefore $\Sigma = S^2$) or so that after gluing up edges, Σ has only one vertex.*

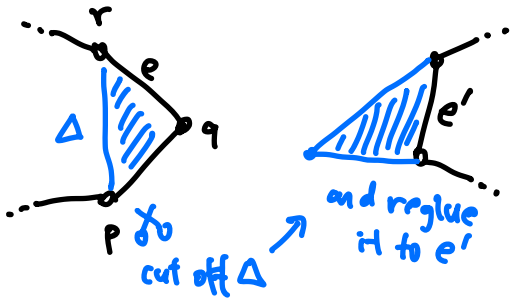
Proof. The gluing determines a surjective (typically many to one) map from the vertices of P to the vertices of Σ. For each vertex v of Σ the *multiplicity* is the number of vertices of P that map to it by the gluing. Order the vertices of Σ $v_1, v_2, \cdots, v_k$ in decreasing order of multiplicity.

If any vertex v_i has multiplicity 1 this means there is a unique vertex w_i of P that maps to it under the gluing. But this means that the two edges of P adjacent to w_i are glued to each other, and either P has only two edges, or else we can just glue these two edges together to get a simpler polygon P' with two fewer edges than P.

If no vertex v_i has multiplicity 1 then no two adjacent edges of P are glued together. Suppose there are at least two vertices in Σ. Choose three consecutive vertices p, q, r in



∂P which map to v_1, v_i, v_j respectively where $i \neq 1$ but we are indifferent about whether $j = 1$ or not. Let e be the edge qr and let e' be the edge of P that is glued to e. We may cut off a triangle Δ of P spanned by p, q, r and attach this triangle to P by gluing e to e'. This produces a new polygon P' with the same number of edges as P.



The net result of this operation is to reduce the multiplicity of v_i by 1 and to increase the multiplicity of v_1 by 1. Thus after finitely many operations of this kind we may arrange for some vertex of Σ to have multiplicity 1, and then we may simplify P as above. The lemma follows by induction. □

If $\Sigma = S^2$ we are done. Otherwise we may perform the modification of P as in Lemma 2.14 so that the gluing produces only one vertex in Σ. For such a gluing we have

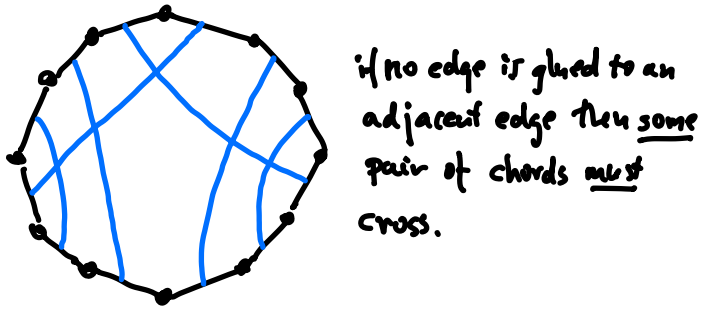
$$\chi(\Sigma) = 2 - \frac{1}{2}\#\text{edges of } P$$

In particular either $\Sigma = 2$ or $\chi(\Sigma) < 2$. We claim that in the latter case that we may write $\Sigma = \Sigma' \# T$ for some closed oriented surface Σ'. Since $\chi(\Sigma') = \chi(\Sigma) + 2$ and (as we have just shown) $\chi(\Sigma') \leq 2$, it will follow by induction that Σ is obtained from S^2 by connect summing g times with T for some non-negative integer g. This will prove the theorem.

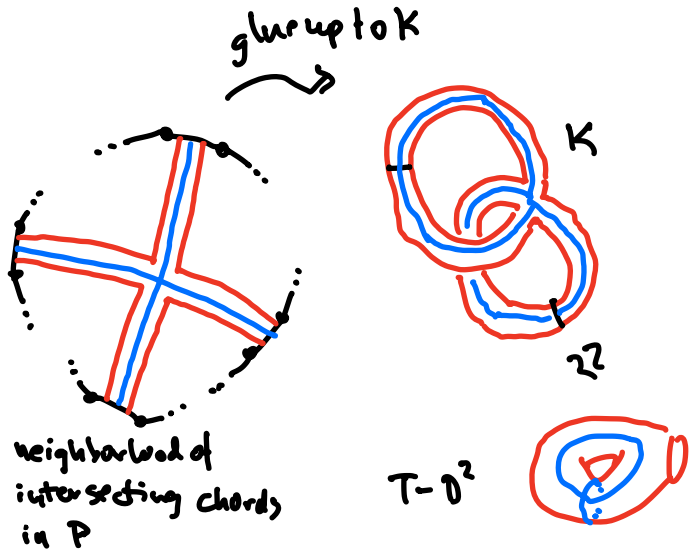
Let's prove the claim. We somehow need to find a way to exhibit Σ as $\Sigma' \# T$. This is evidently the same as finding a compact subsurface K of Σ which is diffeomorphic to T with a disk removed. Where shall we find such a K? Since after gluing there is only one vertex in Σ, it follows that we never glue adjacent edges of P. Let's realize P geometrically as a convex regular polygon in the plane, and for each pair of edges e and e' which are glued, let's draw a straight chord c_e across P joining the midpoints of e and e'.

Suppose these chords were all disjoint. Then we could find an 'innermost' chord — i.e. some chord c_e which splits P into two pieces $P^{\pm}$, one of which (say P^+) contains no other chord. But this is impossible: since the edges e and e' are not adjacent, both P^+ and P^- must contain at least one entire edge, which means they must contain the chord that

lands at that edge. It follows that the chords are *not* all disjoint, and therefore (since they are straight line segments in a convex polygon) there are chords e_1 and e_2 that intersect transversely in exactly one point.



Each chord is glued up to an embedded circle in Σ, and a neighborhood of a chord is glued up to an embedded annulus. Let K be the union of these two annular neighborhoods of e_1 and e_2. Then K is topologically a torus with a disk removed!



Thus we have proved the claim, and therefore $\Sigma = \Sigma' \# T$ for some new surface Σ' with $\chi(\Sigma') = \chi(\Sigma) + 2$. Since we have already shown the Euler characteristic of any closed oriented connected surface is ≤ 2 with equality if and only if it is the sphere, the theorem follows. □

It is worth examining the logic of the proof. Once we have shown that Σ is obtained by gluing up the edges of a polygon P in pairs we could skip directly to the last step and argue that either some pair of adjacent edges are identified (in which case either $\Sigma = S^2$ or P can be simplified, reducing the number of edges by 2) or there are a pair of chords that intersect transversely, and therefore $\Sigma = \Sigma' \# T^2$ for some Σ'. In other words: either $\Sigma = S^2$ or Σ is obtained from some other surface by connect sum with a torus. Why doesn't this complete the proof? The trouble is that it does not rule out the possibility

that we can keep writing

$$\Sigma = \Sigma' \# T^2 = \Sigma'' \# T^2 \# T^2 = \Sigma''' \# T^2 \# T^2 \# T^2 = \cdots$$

indefinitely in a process that never terminates. To show that this process terminates we must either be careful about keeping track of the complexity of some polygon P' that glues up to Σ', relating it to that of P (which is rather fiddly), or to show *a priori* (e.g. as in Lemma 2.14) that $\chi(\Sigma) \leq 2$ for any closed oriented connected surface.

Corollary 2.15. *There are only finitely many knots with crossing number n, for any non-negative integer n.*

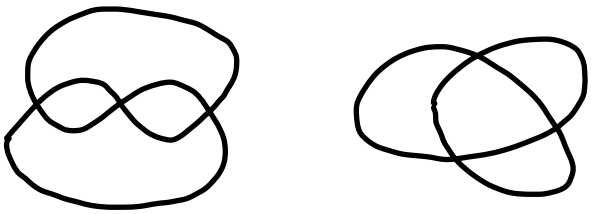
Proof. This is because there are only finitely many *diagrams* (up to topological equivalence) for any n. Suppose D is a diagram with $n > 0$ crossings which we think of as drawn on S^2. Ignoring the data of the crossings, the diagram can be thought of as an immersed circle in the plane, which in turn is a graph Γ with n 4-valent vertices and therefore $2n$ edges. A thickened neighborhood of Γ is a compact oriented surface Σ with $\chi = -n$ (as we shall see in the next exercise) and b boundary components. Each boundary component is a circle, and therefore by the Jordan Curve Theorem bounds a disk (in S^2) in the complement of Σ. In particular, $b = 2 + n$. Each complementary disk can be thought of as a polygon with edges on Γ, and each edge of Γ is obtained by gluing two edges of two complementary polygons. Thus we may reconstruct the diagram topologically (up to the ambiguity of the crossings) from b polygons with $4n$ edges in total by gluing the edges in pairs. Each crossing may be resolved in two possible ways. Thus there is a (computable, but rather messy and not very good) bound on the number of possibilities in terms of n. □

Exercise 2.16. Suppose Γ is a graph embedded in a surface. Let Σ be a thickened neighborhood of Γ. Show that $\chi(\Sigma) = \#\text{vertices}(\Gamma) - \#\text{edges}(\Gamma)$.

Corollary 2.17. *There are no knots with crossing number 1 or 2. The only knots with crossing number 3 are the right handed trefoil and the left handed trefoil (its mirror image).*

Proof. One may enumerate the diagrams with one or two crossings; each of them may have some crossing eliminated by a Reidemeister 1 move; thus they all represent unknots and their crossing number is 0 after all.

If D is a diagram with 3 crossings there are 5 complementary polygons with a total of 12 edges. A polygon with one edge (a 'monogon') may be removed by a Reidemeister 1 move. Thus if D is of minimal complexity each complementary polygon has at least 2 edges and at most 3 (since there are only 3 crossings). Thus the only possibility is that there are two triangles and three bigons.



There are two possible arrangements of these polygons (actually they are equivalent in S^2) and for each arrangement, there are eight ways to choose the three crossings. For each

arrangement, two choices of crossings give rise to the right handed and left handed trefoil respectively, and every other choice gives rise to a diagram that may be simplified by a Reidemeister 2 move. We have already seen that the crossing number for either trefoil is 3. □

The right handed and left handed trefoil are in fact different knots, though we shall not be able to prove this for some time yet.

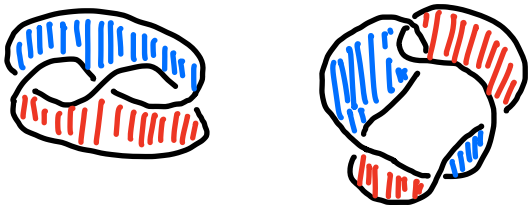
3. Seifert surfaces and homology

3.1. **Genus.** We begin with a definition:

Definition 3.1 (Seifert surface)**.** If K is a knot, the *genus* of K (denoted $g(K)$) is the least genus of a compact oriented connected surface Σ embedded in $\mathbb{R}^3$ whose boundary is (a representative of) K.

Any compact oriented connected surface Σ embedded in $\mathbb{R}^3$ with boundary (a representative of) K is called a *Seifert surface* for K. If C is a representative of K which is equal to $\partial\Sigma$ for some Seifert surface Σ, and C' is equivalent to C, then by Exercise 2.2 the ambient isotopy taking C to C' carries Σ to another Seifert surface Σ' in $\mathbb{R}^3$, topologically equivalent to Σ, whose boundary is C'. Thus the genus may be computed (in principle) from any representative of K.

On the other hand we have no *a priori* reason to believe that such a surface exists at all, and in fact we shall not rigorously prove this until we give Seifert's algorithm in § 3.2. For now let us take it on faith that $g(K)$ is a well-defined non-negative integer for all knots K.



The surfaces are oriented, hence two sided; the two sides are indicated by different colors. Both surfaces have $\chi = -1$ and are therefore of genus 1. The surface on the left bounds a left handed trefoil; a genus 1 Seifert surface for a right handed trefoil may be obtained by mirror image.

Exercise 3.2. Show that both surfaces in this figure have genus 1. What is the knot on the right?

Proposition 3.3. *For any knot K the genus $g(K)$ is zero if and only if K is the unknot.*

Proof. A genus 0 surface with one boundary component is a disk; thus this is a restatement of Proposition 2.1. □

An upper bound on $g(K)$ may be obtained by exhibiting *any* Seifert surface for K. Hence we obtain the following corollary:

Corollary 3.4. *The genus of the right (or left) handed trefoil is 1.*

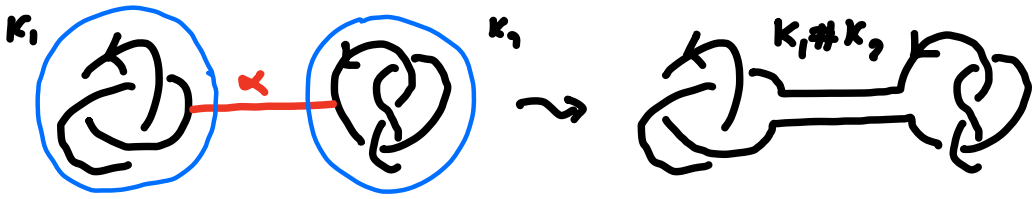
The genus of the figure 8 knot is also 1. We have already proved the upper bound though you may not have noticed it; a lower bound will follow once we (eventually) prove that the figure 8 knot is different from the unknot.

Although genus is in general rather difficult to compute, it does behave very well under an important operation on knots called *connect sum*. Actually, this operation is defined on *oriented* knots, and is defined as follows.

Definition 3.5 (Connect sum)**.** Let K_1 and K_2 be oriented knots. Put representatives of them in disjoint 3-balls B_1 and B_2 in S^3 and join them by an embedded arc α which intersects the knots only at its endpoints and intersects the boundary spheres of each of the B_is transversely in only one point.

Thicken α to a thin rectangle R with two thin edges on the K_i oriented compatibly with the orientation of the knots. Let ∂R^- be the interiors of these two thin edges of ∂R that lie in the K_i, and ∂R^+ the other two edges which run between them. Then $K_1 \# K_2$ is the (oriented!) knot formed by

$$K_1 \# K_2 = \big((K_1 \cup K_2) - \partial R^-\big) \cup \partial R^+$$



Exercise 3.6. Show that the oriented knot type $K_1 \# K_2$ does not depend on the choice of arc α or how it is thickened to a rectangle. Deduce that the operation $\#$ gives rise to an associative, commutative operation on oriented knot types, with the unknot as an identity element.

(Hint: if α appears knotted, shrink down one of the balls B_1 to almost a point so that α looks like it has a free end. Using this free end, 'unknot' the part of α outside B_1. Then expand B_1 and shrink down B_2 and 'unknot' the part of α outside B_2.)

Magicians like to do a trick where they tie two knots together in a rope, then pull the rope tight so that the knots 'cancel' and vanish. This is impossible! If we join up the free ends of the rope, the result of tieing two knots K_1 and K_2 side by side is to produce $K_1 \# K_2$. So we are trying to find nontrivial knots K_1 and K_2 for which the connect sum is the unknot. That this cannot be done follows from the:

Theorem 3.7 (Additivity of genus)**.** *Let K_1 and K_2 be two oriented knots. Then*

$$g(K_1 \# K_2) = g(K_1) + g(K_2)$$

In particular, $K_1 \# K_2$ is the unknot if and only if both K_1 and K_2 are.

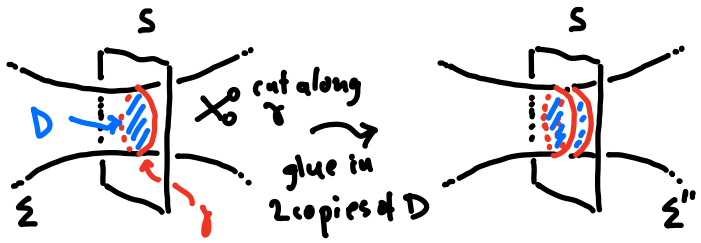
Proof. Let K_1 and K_2 be contained in disjoint 3-balls B_1, B_2. For each i let Σ_i be a minimal genus Seifert surface for K_i. We may radially shrink Σ_i (and K_i) until it is entirely contained in B_i, and then connect up $\partial\Sigma_1$ to $\partial\Sigma_2$ by a thin rectangle R with core

an arc α running between the two knots. The surface $\Sigma := \Sigma_1 \cup R \cup \Sigma_2$ is compact, oriented, connected and has $\partial\Sigma = K_1 \# K_2$. Furthermore, $g(\Sigma) = g(\Sigma_1) + g(\Sigma_2)$. It follows that $g(K_1 \# K_2) \leq g(K_1) + g(K_2)$.

Conversely, let's suppose Σ is a least genus Seifert surface for $K_1 \# K_2$, and let $S = \partial B_1$ be the boundary of the 3-ball that contains K_1 (for concreteness). Wiggle Σ so that it is *transverse* to S; i.e. at every point of intersection the tangent planes to S and to Σ are not equal. Then the intersection $S \cap \Sigma$ is a smooth compact 1-manifold X, with boundary on the intersection $S \cap \partial\Sigma$. Now, S intersects the arc α at only one point, and therefore $S \cap \partial\Sigma$ consists of two points. It follows that X contains a single compact arc β joining these two points, and every other component of X is a circle.

Suppose X does contain some circle component. Then it contains some circle component γ which is *innermost* on S — i.e. which bounds a disk D on one side of $S - \gamma$ that does not contain any other component of X (we are implicitly using the Jordan Curve Theorem here).

How does γ look with respect to Σ? We may cut Σ along γ to produce a new surface Σ' with two new boundary components γ^+ and γ^- (corresponding to the two sides of a neighborhood of γ in Σ) and then sew in two disjoint disks D^+ and D^- along these boundary components obtained by pushing D slightly to either side. This produces a new compact, oriented embedded surface Σ'' with $\partial\Sigma'' = \partial\Sigma = K_1 \# K_2$ and $\chi(\Sigma'') = \chi(\Sigma) + 2$.



Since each component of Σ' must contain at least one of $\gamma^\pm$ it follows that Σ', and therefore also Σ'', has at most two components. If Σ'' were connected it would be a Seifert surface for $K_1 \# K_2$ with $g(\Sigma'') = g(\Sigma) - 1$, contrary to the fact that Σ is a Seifert surface of least genus. Thus Σ'' must have exactly two components Σ^* and F, where Σ^* is the unique component with boundary. Since F is closed, oriented and connected, and since $\chi(\Sigma^*) + \chi(F) = \chi(\Sigma) + 2$ it follows from the minimality of Σ that F is a sphere, and $g(\Sigma^*) = g(\Sigma)$. Thus we have found a *new* Seifert surface Σ^* for $K_1 \# K_2$ with minimal genus, but *with one fewer circle of intersection with* S.

Thus, after finitely many modifications of this kind, we may obtain a new Seifert surface for $K_1 \# K_2$ which we relabel Σ, that intersects S in a single compact arc β. Cutting Σ along β gives two disjoint surfaces Σ_1 and Σ_2 that evidently are Seifert surfaces for (representatives of) K_1 and K_2 respectively, and satisfy $g(\Sigma_1) + g(\Sigma_2) = g(\Sigma)$. It follows that $g(K_1) + g(K_2) \leq g(K_1 \# K_2)$. Combining this with our earlier inequality, the theorem follows. □

Corollary 3.8. *There are infinitely many knots.*

Proof. The result of connect summing a trefoil with itself n times gives rise to a knot with genus n. □

Let us formalize the procedure by which we obtained Σ'' from Σ.

Definition 3.9. Let Σ be a compact oriented surface in $\mathbb{R}^3$ and let γ be an embedded loop in Σ that bounds a properly embedded disk D with interior in $\mathbb{R}^3 - \Sigma$. Let Σ'' be the compact oriented surface in $\mathbb{R}^3$ with $\partial\Sigma'' = \partial\Sigma$ obtained by cutting out a neighborhood of γ and gluing in two parallel disjoint copies of D to the resulting boundary components. Then we say D is a *compressing disk* and Σ'' is obtained from Σ by *compression*.

Any surface may be compressed: if γ is a loop in Σ bounding a disk $E \subset \Sigma$ then we may push the interior of E off Σ to the positive side (say) to obtain a compressing disk D with interior in $\mathbb{R}^3 - \Sigma$ and with $\partial D = \partial E$. One component of Σ is (a slight perturbation of) the 2-sphere $D \cup E$, and the other component has the same genus as Σ. A surface Σ is said to be *incompressible* if for every compressing disk D the boundary circle $\gamma = \partial D$ bounds a disk E in Σ, and to be *compressible* otherwise.

We may reformulate a key argument from Theorem 3.7 in terms of this language:

Proposition 3.10. *Let Σ be a minimal genus Seifert surface for a knot K. Then Σ is incompressible.*

Proof. Suppose Σ is compressible, and D is a compressing disk for which $\gamma = \partial D$ does not bound a disk in Σ, and let Σ'' be the result of compression. As in the proof of Theorem 3.7 we have $\chi(\Sigma'') = \chi(\Sigma) + 2$ so either Σ'' is connected, in which case it is a Seifert surface of genus less than Σ (contrary to hypothesis) or Σ'' has two components Σ^* and F where Σ^* is a Seifert surface and F is closed, oriented and connected. Since $g(\Sigma) \leq g(\Sigma^*)$ it follows that $g(\Sigma) = g(\Sigma^*)$ and $\chi(F) = 2$ so that F is a sphere. By the Jordan Curve Theorem, this sphere is decomposed by (a perturbation of) γ into two disks; one is (a perturbation of) D, and the other is (a perturbation of) a subsurface E of Σ bounded by γ, contrary to the choice of D. It follows that Σ is not compressible after all. □

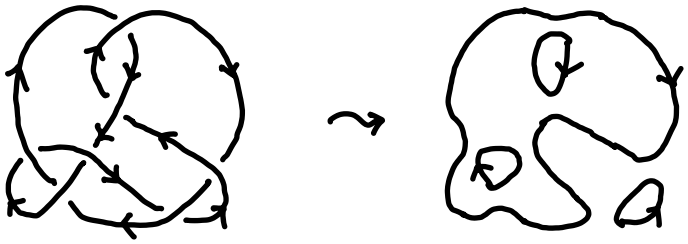
We close this subsection with a parenthetical remark that a cheap way to obtain Corollary 3.8 without appealing to genus is to use 3-colorings. If K_1 and K_2 are knots represented by disjoint diagrams D_1 and D_2 in $\mathbb{R}^2$, and α is an embedded arc in $\mathbb{R}^2$ joining segments of D_1 and D_2 without creating any new crossings, then for any fixed 3-coloring of D_1, the colorings of the segments at the endpoints of α agree for exactly $1/3$ of the colorings of D_2. It follows that $\tau_3(K_1 \# K_2) \geq \tau_3(K_1) \cdot \tau_3(K_2)/3$. In particular, if K is any knot and K' is any other knot with a non-monochromatic 3-coloring (for instance a trefoil) then $\tau_3(K \# K') > \tau_3(K)$.

3.2. **Seifert's algorithm.** Every knot admits a Seifert surface. We shall give a constructive proof of this due to Seifert, in the form of an algorithm that associates an oriented surface to any oriented knot diagram.

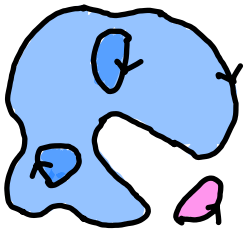
The first step is to take the oriented diagram D and *resolve the crossings*, replacing a neighborhood of each crossing with a pair of disjoint oriented arcs in the unique manner compatible with the orientation.



The result is to replace the original oriented diagram D with a family of oriented disjoint circles in the plane.



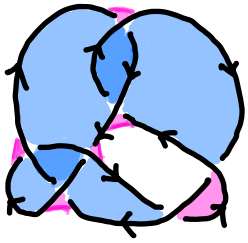
Each circle in the plane bounds a unique disk region by the Jordan Curve Theorem. We build a surface which is the disjoint union of copies of these disk regions, oriented compatibly with the orientations of the circles, so that 'positively' oriented disks have a boundary circle that winds anticlockwise around them, and 'negatively' oriented disks have a boundary circle that winds clockwise around them. Some of these circles might be nested, in which case we put innermost disks 'in front' of the others.



Finally we glue back a twisted disk for each original crossing, connecting up the disks to form a Seifert surface Σ for K (when the two disks being connected are nested, the twisted disk is 'folded'):



For example, the Seifert surface for the diagram of the knot 7_6 from Rolfsen's tables [8] is shown below:



What is the genus of Σ? We may estimate an upper bound as follows. After resolving all the crossings our surface consists of a finite union of disks; each disk has $\chi = 1$ so this surface has $\chi = n$ where n is the number of circle components. Attaching each twisted disk reduces the Euler characteristic by 1, and we attach $c(D)$ disks, one for each crossing. It follows that $\chi(\Sigma) \geq 1 - c(D)$. Since Σ has one boundary component, $\chi(\Sigma) = 1 - 2g(\Sigma)$ and therefore $g(\Sigma) \leq c(D)/2$ and $g(K) \leq c(K)/2$.

In our example $n = 4$ and $c(D) = 7$ so $\chi(\Sigma) = -3$ and $g(\Sigma) = 2$. In fact, this is the genus of the knot 7_6, though we do not yet have the tools to prove it.

3.3. **Linking number.** A *link* is an equivalence class of smooth (or polygonal) disjoint embedded circles in $\mathbb{R}^2$ up to smooth (or polygonal) equivalence. A knot is a special case of a link with one component. For each n the *n-component unlink* is represented by n disjoint unknots, each unknot embedded in a separate 3-ball disjoint from the others.

Links, just like knots, are represented by diagrams, and two diagrams represent the same link if they are related by Reidemeister moves. A link is *oriented* if each of its components is separately oriented; thus a link with n components admits 2^n distinct orientations.

Definition 3.11 (Linking number)**.** Let L be an oriented link with 2 components K_1 and K_2. The *linking number* of the two components, denoted $\ell(K_1, K_2)$, may be computed from any diagram by counting the number of *positive* overcrossings of K_1 over K_2 minus the number of *negative* overcrossings of K_1 over K_2:



Proposition 3.12. *Linking number is an* invariant*; i.e. it does not depend on the choice of diagram representing a link. Furthermore it is symmetric: $\ell(K_1, K_2) = \ell(K_2, K_1)$ and changes sign if the orientation of one of the components of a link is reversed.*

Proof. Changing the orientation of one of the components changes a positive overcrossing into a negative overcrossing and vice versa, proving the third claim.

To show that linking number is an invariant, we must just check that it does not change under a Reidemeister move. Reidemeister 1 creates (or removes) a crossing, but this crossing necessarily involves a single component, and can't contribute to linking number. Thus we just need to check invariant under Reidemeister 2 and Reidemeister 3, which is straightforward and left as an exercise.

Once we know that linking number is an invariant, consider what happens to a diagram when we take a link and 'rotate it' through 180 degrees about the horizontal axis contained in the plane of the projection. The effect is to 'reflect' the diagram in this axis and at the same time to change overcrossings to undercrossings and vice versa. Thus, under this transformation, each positive crossing of K_1 over K_2 is changed into a positive crossing of K_2 over K_1 and vice versa. The identity $\ell(K_1, K_2) = \ell(K_2, K_1)$ follows. □

We would now like to relate linking number to Seifert surfaces. Let L be an oriented link with 2 components K' and K. Fix disjoint oriented representative circles C' and C and let Σ be any Seifert surface with boundary C. Wiggle C' to be transverse to Σ.

Define the *signed intersection number* of C' and Σ to be the number of intersections of C' with Σ counted with a positive sign when C' crosses through Σ from the negative to the positive side, and counted with a negative sign when C' crosses through Σ from the positive to the negative side.

Theorem 3.13 (Linking number is intersection number)**.** *With notation as above, the signed intersection number of C' and Σ is equal to the linking number $\ell(K', K)$. In particular, it does not depend on the choice of representatives C', C or the choice of Seifert surface Σ.*

The proof of this theorem will follow after we develop some machinery. Roughly speaking, we establish the following three facts:

(1) we shall associate to any oriented embedded circle C in $\mathbb{R}^3$ the free Abelian group of *additive invariants* which consists of functions from oriented circles in $\mathbb{R}^3 - C$ to $\mathbb{Z}$, and which satisfies a certain additivity axiom (to be described below);
(2) we shall show that linking number with C, and signed intersection number with Σ are both additive invariants; and
(3) we shall show that as an abstract group, the group of additive invariants is isomorphic to $\mathbb{Z}$, generated by the function that takes the value 1 on a particular knot (an *oriented meridian* for C).

The proof will then be completed by showing that $\ell(\cdot, C)$ and signed intersection number with Σ are both equal to 1 for a meridian, and therefore they are equal for any embedded oriented circle C' in $\mathbb{R}^3 - C$.

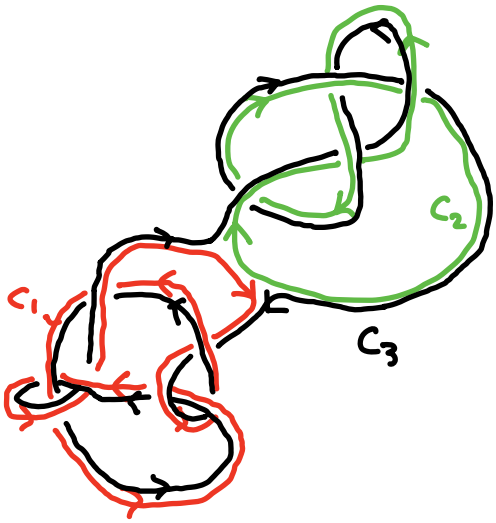
Let us now define the Abelian group of additive invariants. Fix a representative C of the knot K. The group consists of functions f from (smooth or polygonal) circles C' in $\mathbb{R}^3 - C$ to $\mathbb{Z}$ which are

(1) **invariant:** a function f is *invariant* if it is constant on (smooth or polygonal) equivalence classes of knots in $\mathbb{R}^3 - C$; and
(2) **additive:** for any two disjoint oriented circles C_1, C_2 which are almost parallel but oppositely oriented along a pair of segments, we may obtain a new oriented knot C_3 by cutting out the pair of segments and inserting a new pair of oriented segments as so:



(Let us refer to this in the sequel as the 'rewiring' move). A function f is *additive* if for any triple of knots C_1, C_2, C_3 related as above we have $f(C_1) + f(C_2) = f(C_3)$.

An example of a triple of knots C_1, C_2, C_3 related by the rewiring move described above is illustrated in the following figure. For any additive invariant $f \in A$ one must have $f(C_1) + f(C_2) = f(C_3)$.



We now prove two lemmas, which show that $\ell(\cdot, C)$ and signed intersection number with Σ are both additive invariants.

Lemma 3.14. *Linking number with C is an additive invariant.*

Proof. We have already shown that linking number (as computed e.g. from a diagram) is an invariant since it does not change under Reidemeister moves. If C_1, C_2, C_3 are three circles related by the rewiring move, then after replacing the circles by equivalent circles we may assume that the rewiring takes place somewhere in the diagram away from crossings with C. In particular, $C_1 \cup C_2$ has the same overcrossings of C (with the same signs) as C_3, so $\ell(C_1, C) + \ell(C_2, C) = \ell(C_3, C)$. □

Lemma 3.15. *Signed intersection number with Σ is an additive invariant.*

Proof. Let us first show that signed intersection number is an invariant. Let C' and C'' be two oriented circles in $\mathbb{R}^3 - C$ that intersect Σ transversely. To say that C' and C'' represent the same knot type in $\mathbb{R}^3 - C$ means that there is a smooth map $g : S^1 \times [0,1] \to \mathbb{R}^3 - C$ for which the (oriented) image $g(S^1 \times 0) = C'$ and $g(S^1 \times 1) = C''$. We may wiggle the map g so that it intersects Σ transversely; in other words, we may arrange that $g^{-1}(\Sigma)$ is a system X of embedded circles and properly embedded arcs in $S^1 \times [0,1]$. The orientations on Σ and on $S^1 \times [0,1]$ determine orientations on the components of X; there are several ways to do this, and we leave it as an exercise to the reader.

As the oriented boundary of the oriented surface $S^1 \times [0,1]$ the orientations on $S^1 \times 0$ and on $S^1 \times 1$ respectively *disagree* and *agree* with the orientations on C' and C'' induced by g. It follows that the signed intersection number of C'' with Σ is the signed intersection number of X with $S^1 \times 1$, whereas the signed intersection number of C' with Σ is the *negative* of the signed intersection number of X with $S^1 \times 0$. Thus to prove that the signed intersection numbers with Σ are the same for C' and C'' it suffices to show that the signed intersection number of X with $\partial(S^1 \times [0,1])$ is zero. But the circle components of X do not intersect $\partial(S^1 \times [0,1])$ at all, whereas each proper arc intersects it twice, with opposite sign. The claim follows.

Once we know that signed intersection number is an invariant, it is easy to show that it is additive. Let C_1, C_2, C_3 be three circles related by the rewiring move. After replacing these circles by equivalent circles we may assume (as in the proof of Lemma 3.14) that the rewiring takes place away from Σ. In particular, $C_1 \cup C_2$ has the same intersections of Σ (with the same signs) as C_3, and we are done. □

Exercise 3.16. Let A and B be two oriented surfaces in $\mathbb{R}^3$ that intersect transversely in a system X of arcs and circles. Show how the orientations on A and B determine orientations on the components of X.

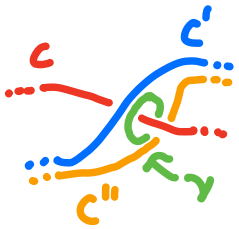
Let C be an oriented embedded circle in $\mathbb{R}^3$. A *meridian* μ for C is a tiny unknotted oriented circle μ that bounds a tiny flat disk that intersects C transversely in one point, as in the figure.



Proposition 3.17. *Let C be an oriented embedded circle in $\mathbb{R}^3$. Let ϕ be any additive invariant of oriented circles in $\mathbb{R}^3 - C$. Then ϕ is determined by its value on any meridian μ for C.*

Proof. It is easy to show directly from the definition that any additive invariant must vanish on an unknot (i.e. a circle that bounds an embedded disk in $\mathbb{R}^3 - C$), and that changing the orientation of an oriented circle multiplies the value of any additive invariant by -1.

Let C' be any oriented circle in $\mathbb{R}^3 - C$ and consider a diagram for C and C'. Fix an overcrossing of C' over C (if any) and let C'' be the oriented circle obtained from C' by changing that overcrossing to an undercrossing. Then C' and C'' are related by a rewiring move in which the third circle is a meridian ν, so that $\phi(C') = \phi(\nu) + \phi(C'')$.



Since ν is a meridian, by sliding it around C we may exhibit an equivalence to μ which is either orientation-preserving if the overcrossing was positive, or orientation-reversing if the overcrossing was negative. Thus $\phi(C') = \phi(C'') \pm \phi(\mu)$, where the sign of the $\phi(\mu)$ term is the sign of the overcrossing. By induction we obtain $\phi(C') = \phi(C^*) + \sum \pm \phi(\mu)$ where the sum is taken over all overcrossings of C' over C, and where C^* has no overcrossings over C. But if C^* has no overcrossings over C we may perform a sequence of Reidemeister moves (pushing C^* to one side from below) so that C^* has no crossings with C at all.

If C^* and C^{**} are related by a self-crossing change of C^* then arguing as above $\phi(C^*) = \phi(\lambda) + \phi(C^{**})$ where now λ is a meridian *for* C^*. But a meridian for C^* is unknotted in $\mathbb{R}^3 - C$ and therefore $\phi(\lambda) = 0$ and hence $\phi(C^*) = \phi(C^{**})$. In other words, we may change

all the self-crossings of C^* without affecting ϕ. But then we can change C^* to an unknot in $\mathbb{R}^3 - C$ and therefore conclude that $\phi(C^*) = 0$. Hence $\phi(C') = \sum \pm\phi(\mu)$ and the value of ϕ on C' is completely determined by the value on μ. □

Exercise 3.18. Let C be an oriented circle and let ϕ be an additive invariant on circles in $\mathbb{R}^3 - C$. Show that ϕ vanishes on unknots (that bound disks in $\mathbb{R}^3 - C$), and if C' and C'' are the same circle with opposite orientations, then $\phi(C'') = -\phi(C')$. (Hint: show that there is a rewiring move in which all three circles are unknots; and if C' and C'' are the same circle with opposite orientations, show there is a rewiring move involving C', C'' and an unknot).

At this point the proof of Theorem 3.13 follows almost immediately:

Proof. To prove the theorem it suffices to prove it when $C' = \mu$, an oriented meridian for C. An oriented meridian has exactly one positive overcrossing in a diagram, so $\ell(\mu, C) = 1$. On the other hand, an oriented meridian also has exactly one positive intersection with any Seifert surface Σ. The proof follows. □



4. The knot group

4.1. **A return to coloring.** Recall that we defined a 3-coloring of a diagram D to be an assignment of one of three colors to each of the segments of D so that at each crossing, the colors assigned to the three segments that meet at the crossing are either all the same or all different.

If we are drawing an actual diagram on paper, then we know what we mean by using different colors to draw the segments of the diagram. But if we want to abstract the mathematical essence of what a coloring 'is', then it is easy to see that it is nothing more than a map of sets: a map from the set of segments to some fixed 3 element set, and subject to certain local conditions. As Tolstoy might say, all 3 element sets are alike — they are all interchangable *qua sets* — but if we consider 3 element sets *endowed with extra structure* then it becomes possible to express the coloring conditions (all colors the same or all colors different) in more meaningful terms.

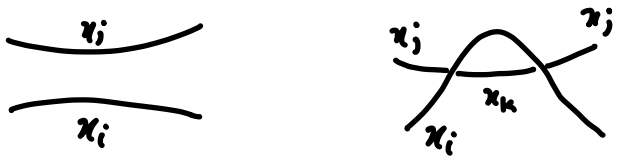
Let's take the three 'colors' to be elements of the Abelian group $\mathbb{Z}/3\mathbb{Z}$ of integers modulo 3; in other words, each segment e_i of the diagram D gets assigned an element $x_i \in \mathbb{Z}/3\mathbb{Z}$. Then the coloring condition at each crossing (involving segments e_i, e_j, e_k) becomes the *algebraic* condition $x_i + x_j + x_k = 0 \bmod 3$. Since $\mathbb{Z}/3\mathbb{Z}$ is a field, and the coloring conditions are linear, the set of 3-colorings gets the structure of a (finite dimensional) *vector space* over $\mathbb{Z}/3\mathbb{Z}$. In particular, the cardinality of this set is a power of 3.

Once we frame things in this way we can start to fiddle with some of the parameters and see what happens. Let's try to replace the field $\mathbb{Z}/3\mathbb{Z}$ with another field $\mathbb{Z}/p\mathbb{Z}$ for a different prime p. Suppose e is a segment in a diagram D colored by some $x \in \mathbb{Z}/p\mathbb{Z}$. If

we perform Reidemeister 1 in the middle of the segment e we get a new diagram D' with one new crossing at which three segments meet. In any coloring of D' which agrees with that of D outside the region in which we have performed the move, the three segments meeting at the new crossing must all have the same coloring, and therefore we get the equation $3x = 0$. Since x was arbitrary this implies that $x = 0$ unless $p = 3$. Our attempt to generalize 3-colorings to p-colorings has fallen at the first hurdle.

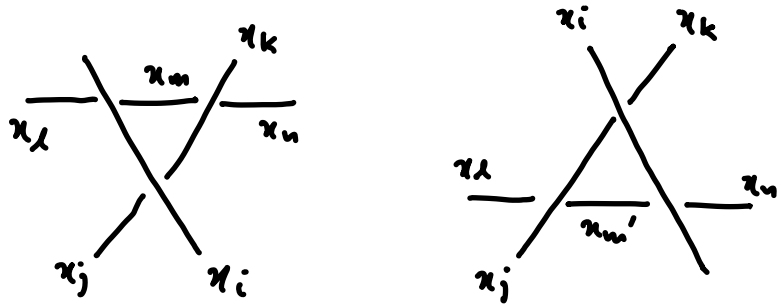
Let's take a step back. We changed the parameter 3 to the parameter p but we kept the same algebraic equation $x_i + x_j + x_k = 0$. Are we free to change the equation? The equation $x_i + x_j + x_k = 0$ is symmetric in the three variables corresponding to the three segments meeting at a crossing. But at a crossing the diagram is *not* symmetric in the three segments: one segment (e.g. e_i) crosses over, while the other two (e_j, e_k) may be interchanged by a (local) 180 degree rotation. Thus we are compelled to look for equations which are symmetric in x_j and x_k but not necessarily in all 3 variables. Arguing as above, the only linear equation of this form which is compatible with Reidemeister 1 (up to multiplication by a scalar) is $x_j + x_k = 2x_i$.

Now the situation is much better. Define $\tau_p(D)$ to be the number of functions from segments of D to $\mathbb{Z}/p\mathbb{Z}$ satisfying $x_j + x_k = 2x_i$ at every crossing where x_i corresponds to the overcrossing segment. Let's verify that $\tau_p(D) = \tau_p(D')$ whenever D and D' differ by a single Reidemeister move. We have already considered Reidemeister 1. Reidemeister 2 is indicated in the following figure:



In the second figure we have one new variable x_k. The coloring equations both impose the relation $x_k = 2x_i - x_j$; thus every solution of the coloring equations for the first diagram corresponds to a unique solution of the coloring equations for the second diagram.

In a similar way we can consider the effect of Reidemeister 3:



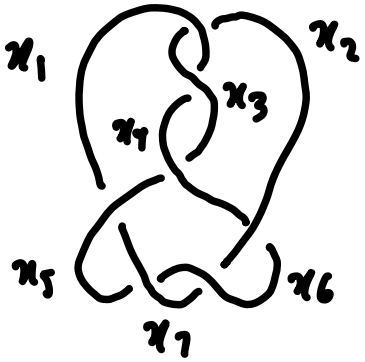
In the diagram on the left the equations $x_l + x_m = 2x_i$ and $x_m + x_n = 2x_k$ give the relation $x_l - x_n = 2(x_i - x_k)$ whereas on the right $x_l + x'_m = 2x_j$ and $x'_m + x_n = 2x_i$ yield $x_l - x_n = 2(x_j - x_i)$. Thus compatibility of the colorings requires $x_i - x_k = x_j - x_i$ or $x_j + x_k = 2x_i$ which is true in either diagram because of the 'common' overcrossing. In other

words: each diagram imposes the same linear conditions on the variables x_i, x_j, x_k, x_l, x_n, and any valid solution to these conditions uniquely determines either $x_m := 2x_i - x_l$ (on the left) or $x'_m := 2x_j - x_l$ (on the right).

We conclude that $\tau_p(D) = \tau_p(D')$ whenever D and D' differ by a single Reidemeister move, and therefore $\tau_p(K)$ is a well-defined invariant of the underlying knot K. This invariant counts the number of elements in a vector space over $\mathbb{Z}/p\mathbb{Z}$ and is therefore a power of p; thus we obtain the same information by tracking the *dimension of the space of p-colorings*. We denote this $n_p(K)$ and observe that $n_p(K) = \log_p \tau_p(K)$.

Example 4.1 ($p = 2$). For $p = 2$ the coloring equations reduce to $x_j = x_k$ whenever segments e_j, e_k meet at an underpass. Evidently this implies that the only valid 2-colorings are monochromatic; i.e. $n_2(K) = 1$ for every K.

Let's consider a specific knot. This figure shows (a diagram of) the knot 7_3 from Rolfsen's tables [8]. There are 7 crossings and hence 7 segments that we color $x_1, \cdots, x_7$ as in the figure. The coloring conditions can be expressed in the form $Ax = 0$ where x is a vector with entries x_i, and where A is a 7×7 matrix whose rows give the linear conditions associated to the 7 crossings.



From the figure we can read off the matrix entries

$$\begin{pmatrix} -2 & 1 & 1 & 0 & 0 & 0 & 0 \\ 1 & 0 & -2 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & -2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & -2 & 0 & 1 \\ 0 & -2 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 & -2 \\ 0 & 1 & 0 & 0 & 0 & -2 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \\ x_7 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

The sum of all column vectors is the zero vector; this is equivalent to the fact that monochrome colorings are always valid, and the analogous fact is true for every diagram of every knot. In this particular case it is also true that the sum of all row vectors is the zero vector, so by Gaussian elimination (i.e. row and column operations) we may reduce to the upper 6×6 block which has determinant 13. In particular, we deduce that $n_p(7_3) = 1$ if $p \neq 13$ and $n_{13}(7_3) = 2$.

It should be clear that a similar situation holds for any knot. We can always express the coloring conditions in the form $Ax = 0$ for A a $c \times c$ matrix (where c is the crossing

number). The columns sum to zero (because monochromatic colorings are always valid), and therefore by row and column operations we can eliminate the last row and column, and reduce to a $(c-1) \times (c-1)$ submatrix with determinant Δ.

Lemma 4.2. *Δ is odd; in particular, it is nonzero.*

Proof. If Δ were even, the rank of the reduction of the coloring matrix A over $\mathbb{Z}/2\mathbb{Z}$ would be at most $c-2$ and therefore the space of solutions over $\mathbb{Z}/2\mathbb{Z}$ would be at least 2 dimensional. In particular, we could find a nonconstant solution over $\mathbb{Z}/2\mathbb{Z}$, contrary to Example 4.1. □

Since Δ is nonzero, we can compute the rank of the kernel of A over $\mathbb{Z}/p\mathbb{Z}$ for any p by counting the numbher of times p occurs as a factor of Δ, and adding 1. We denote this number by $\nu_p(\Delta)$, so that we arrive at the formula $n_p = 1 + \nu_p(\Delta)$. Changing the ordering of the variables will multiply Δ by ± 1. Thus the information contained in all n_p for all primes p is exactly equivalent to knowledge of the positive integer $|\Delta|$; i.e. $|\Delta|$ is a knot invariant called the *determinant* of the knot, which is exactly as powerful as the collection of all knot invariants n_p together.

If K is an *oriented* knot we can push the idea of colorings even further. At an overcrossing where e_i is the overcrossing segment and e_j, e_k are the undercrossing segments we may use the orientation on e_i to distinguish between e_j on the left and e_k on the right. Then we may look for a linear equation in x_i, x_j, x_k which is now no longer symmetric in any pair of variables. Up to scale we may choose coefficients -1 and t for x_k and x_j respectively where t represents a formal variable whose properties we shall examine shortly. Compatibility with Reidemeister 1 forces the coefficient of x_i to be $(1-t)$; thus we are led to the coloring condition

$$tx_j - x_k + (1-t)x_i = 0$$

(note that this specializes to our previous symmetric coloring condition for unoriented knots when $t = -1$).

For each oriented diagram D and for each unknown t we may compute the dimension $n_t(D)$ of the space of colorings as above, and check that $n_t(D) = n_t(D')$ whenever D and D' are related by a Reidemeister diagram; if we are careful this can be done while remaining agnostic about the field in which our colors take values. The only caveat is that we must insist that $t \neq 0$.

Exercise 4.3. Show that if $t \neq 0$ the dimensions $n_t(D) = n_t(D')$ whenever D and D' are oriented diagrams related by a single Reidemeister move.

As before, if D is an oriented diagram with c crossings, the coloring condition can be expressed as a linear matrix equation $Ax = 0$ where now A is a matrix whose entries are in $\{0, -1, t, (1-t)\}$. The columns still sum to 0 so we may eliminate a row and a column by Gaussian elimination, and take the determinant of the remaining $(c-1) \times (c-1)$ block. This determinant is now a *polynomial* in t with *integer coefficients*. This polynomial is called the *Alexander polynomial* of K, and by abuse of notation we write it $\Delta(t)$ or $\Delta_K(t)$ if we want to emphasize that it depends on the underlying knot K. We may recover our previous invariants by specializing to $t = -1$.

Let's look at some examples. For the right handed trefoil we have

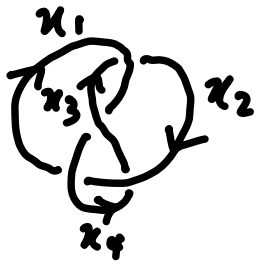


From the figure we can read off the matrix entries

$$\begin{pmatrix} 1-t & t & -1 \\ -1 & 1-t & t \\ t & -1 & 1-t \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

so that $\Delta(t) = t^2 - t + 1$. Note that $\Delta(-1) = 3$ reflecting the existence of a nontrivial 3-coloring for the trefoil.

For the figure 8 knot we have



From the figure we can read off the matrix entries

$$\begin{pmatrix} 1-t & t & -1 & 0 \\ -1 & 0 & 1-t & t \\ -1 & t & 0 & 1-t \\ 0 & 1-t & -1 & t \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \\ 0 \end{pmatrix}$$

so that $\Delta(t) = t^2 - 3t + 1$. Since $\Delta(-1) = 5$ there is a nontrivial 5-coloring for the figure 8 knot (can you find one?)

4.2. **The knot group.** We now show how to associate a *group* to an oriented knot. Recall that a *finite presentation* of a group is given by a finite list of generators $g_1, g_2, \cdots, g_n$ together with a finite list of relations $r_1, r_2, \cdots, r_m$ which are each finite words in the generators and their inverses. If G is the group defined by this presentation we write

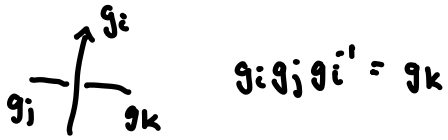
$$G := \langle g_1, \cdots, g_n \mid r_1, \cdots, r_m \rangle$$

Different presentations may give rise to isomorphic groups. There are a small number of operations on group presentations, called *Tietze transformations*, that give rise to isomorphic groups; and conversely any two presentations that give rise to isomorphic groups are related by a finite number of Tietze transformations. These operations are:

(1) adding a new relation r that is a consequence of the other relations; and
(2) adding a new generator g together with a new relator of the form $g = w$ where w is a word in the existing generators g_i and their inverses;

together with their inverses. In the sequel we shall only use the (easy) fact that these transformations give rise to isomorphic groups.

Now let D be an oriented knot diagram. We associate to the diagram a finite presentation of a group by taking one generator g_i for each oriented segment e_i, and at each crossing we add the relation $g_i g_j g_i^{-1} g_k^{-1}$ (which can also be expressed in the form $g_i g_j g_i^{-1} = g_k$).

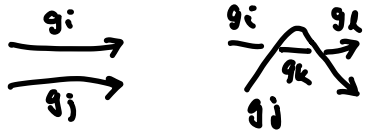


Note that although we need the orientation on the overcrossing segment to decide which undercrossing segment is to the left and which is to the to the right, we do not need the orientation on the undercrossing segments to express the relation. Note too that in the group defined by this presentation, all the generators are *conjugate* to each other; in particular, they pick out a distinguished conjugacy class.

Let's consider the effect of Reidemeister moves.



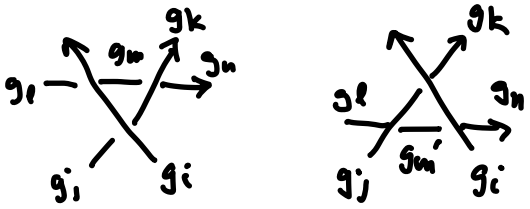
Reidemeister 1 introduces a new generator g_j and a new relation $g_i g_i g_i^{-1} = g_j$ which simplifies to $g_i = g_j$. Thus diagrams related by Reidemeister 1 give rise to presentations related by a Tietze transformation.



Reidemeister 2 introduces two new generators g_k and g_l, and two new relations $g_j g_i g_j^{-1} = g_k$ and $g_j g_l g_j^{-1} = g_k$ which may be rewritten as $g_j^{-1} g_k g_j = g_l$. Thus diagrams related by Reidemeister 2 give rise to presentations related by two Tietze transformations (adding the generator g_k and the first relation; and then adding the second generator g_l and the second rewritten relation).

Finally let's consider Reidemeister 3.

In the left diagram, we have relations $r_1 := g_i g_l g_i^{-1} g_m^{-1}$ and $r_2 := g_k g_m g_k^{-1} g_n^{-1}$. These two relations imply the relation $r' := g_k g_i g_l g_i^{-1} g_k^{-1} g_n^{-1}$. On the other hand, r_1 and r' together imply r_2. We may therefore add the relation r' and remove r_2 by successive Tietze transformations. But then we may rewrite r_1 in the form $g_i g_l g_i^{-1} = g_m$ and use this to eliminate both the relation r_1 and the generator g_m.



Similarly in the right diagram, we have relations $s_1 := g_j g_l g_j^{-1} g_{m'}^{-1}$ and $s_2 := g_i g_{m'} g_i^{-1} g_n^{-1}$. As before these relations imply a new relation $s' := g_i g_j g_l g_j^{-1} g_i^{-1} g_n^{-1}$, and s_1 and s' together imply s_2. We may therefore replace the relation s_2 by s', and then eliminate the generator $g_{m'}$ and the relation s_1.

On the other hand, in both diagrams we have the relation $t : g_i g_j g_i^{-1} g_k^{-1}$ or equivalently $g_i g_j = g_k g_i$. Evidently r' and t together imply s', and conversely s' and t together imply r'. We may therefore (by applying two successive Tietze transformations) replace r' by s' and transform the presentation for the left diagram into the presentation for the right diagram.

In conclusion: diagrams related by Reidemeister moves give rise to isomorphic groups. If one is careful, one can check that the natural isomorphisms associated to Tietze transformations carry the distinguished conjugacy class represented by the generators in one presentation to the distinguished conjugacy class represented by the generators in the other. In particular we can associate to each oriented knot K a group $G(K)$ called the *knot group* together with a distinguished conjugacy class $\gamma(K)$, up to isomorphism.

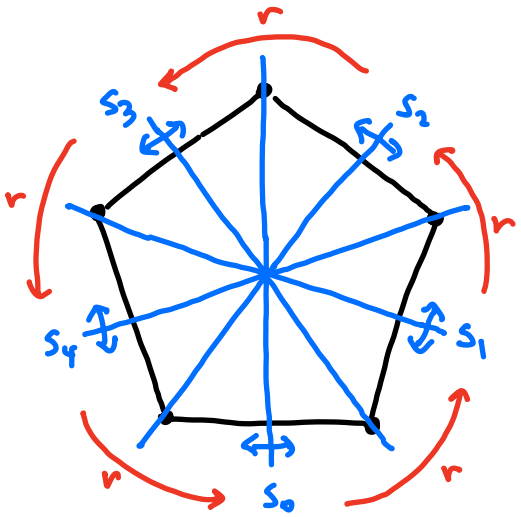
The knot group is an extremely powerful invariant. For instance, $G(K)$ is isomorphic to $\mathbb{Z}$ if and only if K is the unknot. The trouble is that it can be very difficult to decide whether two finite presentations give rise to isomorphic groups. In fact, it turns out that there is *no algorithm* which will always decide, given as input two group presentations, whether they give rise to isomorphic groups. On the other hand, from a group presentation one can at least write down simpler invariants which can be more easily compared.

One easily computed invariant of a group is its *abelianization*, the quotient group obtained by adding as new relations that each pair of generators commutes. From the form of $G(K)$ one sees that in the abelianization G_{ab} the relations in G coming from the crossings simplify to say that a pair of generators are equal. Thus G_{ab} is isomorphic to the (additive) group $\mathbb{Z}$, with the distinguished conjugacy class γ mapping to the generator 1.

A more useful invariant may be obtained by fixing a finite group H and a conjugacy class α and then counting homomorphisms from G to H sending γ into α. As an example, let H be the dihedral group D_p, the group of symmetries of a regular p-gon where p is an odd prime.

Let r be (anticlockwise) rotation through angle $2\pi/p$, so that r has order p. Label the sides 0 through $p-1$ in order, and let s_i be the reflection in the perpendicular bisector of side i. Thus the $2p$ elements of H are r^i and s_i for $i = 0, 1, \cdots, p-1$. Each s_i has order 2, and they are all conjugate to each other: $r^i s_j r^{-i} = s_{i+j}$. They satisfy the relation $s_j s_i = r^{2(j-i)}$ (remember that composition of group elements corresponds to composition of symmetries, so the symmetry gh means first do h and then g).

How many homomorphisms are there from $G(K)$ to D_p sending γ to the conjugacy class of the s_i? Fix a diagram D for K and consider the presentation associated to the diagram. Each oriented edge gives rise to a generator g_i which must map to some s_{x_i} for $x_i \in \mathbb{Z}/p\mathbb{Z}$.



At a crossing the relation $g_i g_j g_i^{-1} g_k^{-1}$ imposes the condition $s_{x_i} s_{x_j} = s_{x_k} s_{x_i}$ or equivalently $2(x_i - x_j) = 2(x_k - x_i)$ which simplifies to $x_j + x_k = 2x_i$. In other words, homomorphisms from G to D_p sending γ to the conjugacy classes of s_i are the same things as p-colorings!

4.3. **Solvable representations and the Alexander Polynomial.** Let H denote the group of 2×2 complex matrices with determinant 1 of the form $\left(\begin{smallmatrix} z & w \\ 0 & z^{-1} \end{smallmatrix}\right)$ for $z \in \mathbb{C}^*$ and for each $t \in \mathbb{C}^*$ and each choice of square root $t^{1/2}$ let $H_{t^{1/2}}$ denote the set of matrices for which $z = t^{1/2}$. We compute:

$$\begin{pmatrix} z & w \\ 0 & z^{-1} \end{pmatrix} \begin{pmatrix} t^{1/2} & \alpha \\ 0 & t^{-1/2} \end{pmatrix} \begin{pmatrix} z^{-1} & -w \\ 0 & z \end{pmatrix} = \begin{pmatrix} t^{1/2} & z^2\alpha + zw(t^{-1/2} - t^{1/2}) \\ 0 & t^{-1/2} \end{pmatrix}$$

It follows that the $H_{t^{1/2}}$ form a single conjugacy class when $t \neq 1$, while each of $H_{\pm 1} - \mathrm{id}$ form a single conjugacy class when $t^{1/2} = \pm 1$.

Fix a knot K, and let's consider representations from $G(K)$ to H sending the conjugacy class of γ into $H_{t^{1/2}}$. When $t = 1$ the set H_1 forms an Abelian *group*, isomorphic to $\mathbb{C}$ with its additive structure, so any representation from $G(K)$ sending the generators into H_1 must map entirely into H_1 and must therefore factor through the abelianization $\mathbb{Z}$, so that every generator actually maps to the *same* matrix in H_1, and the choice of this matrix is arbitrary (a similar story holds for H_{-1}).

Now let's consider the case $t \neq 1$. If we fix a diagram, each generator g_i maps to a matrix for which the upper off-diagonal entry is a complex number x_i. Then the product $g_i g_j g_i^{-1}$ maps to

$$\begin{pmatrix} t^{1/2} & x_i \\ 0 & t^{-1/2} \end{pmatrix} \begin{pmatrix} t^{1/2} & x_j \\ 0 & t^{-1/2} \end{pmatrix} \begin{pmatrix} t^{-1/2} & -x_i \\ 0 & t^{1/2} \end{pmatrix} = \begin{pmatrix} t^{1/2} & tx_j + (1-t)x_i \\ 0 & t^{-1/2} \end{pmatrix}$$

and therefore the relation $g_i g_j g_i^{-1} g_k^{-1}$ is equivalent to the relation $tx_j + (1-t)x_i - x_k = 0$, which is precisely the t-coloring condition for oriented knots! In other words: roots of the Alexander polynomial are the t for which there are nonabelian representations $G \to H$ sending the conjugacy class of the generators into $H_{t^{1/2}}$.

With the aid of some group theory and commutative algebra we can recast the discussion above more directly in terms of the algebraic structure of the group G.

The group H is 2-step solvable; this means that it admits a surjective homomorphism to an Abelian group for which the kernel is also Abelian. The surjective homomorphism $H \to \mathbb{C}^*$ takes a matrix to its upper left entry; and the kernel is the subgroup H_1, which is isomorphic to the *additive* group $\mathbb{C}$.

The knot group G is typically not 2-step solvable (in fact, it is rarely solvable at all). But any homomorphism from G to any 2-step solvable group factors through a maximal 2-step solvable *quotient* of G. Thus the information encoded in the Alexander polynomial may be obtained directly by an analysis of the structure of this 2-step solvable quotient.

We have already seen that there is a maximal Abelian quotient of G, namely $G_{\text{ab}} = \mathbb{Z}$. The kernel is the commutator subgroup, usually denoted G'. Now, G' is typically not Abelian, but it admits a map to its own abelianization $(G')_{\text{ab}}$ with kernel $(G')'$ which is usually denoted $G^{(2)}$. Although in general a normal subgroup of a normal subgroup is not necessarily normal in the bigger group, the subgroup $G^{(2)}$ is normal in G. We may therefore form the 2-step solvable quotient $G/G^{(2)}$ which maps to $\mathbb{Z}$ with kernel the Abelian group $(G')_{\text{ab}}$.

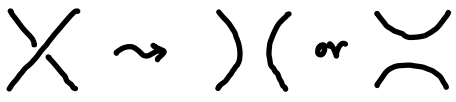
Let's write the Abelian group $(G')_{\text{ab}}$ as A. The generator 1 of $\mathbb{Z}$ is in the image of some $g \in G/G^{(2)}$, and conjugation by g induces an automorphism $t : A \to A$. We may therefore think of A as a $\mathbb{Z}[t, t^{-1}]$-*module*.

It turns out that A is finitely generated as a $\mathbb{Z}[t, t^{-1}]$-module, and that for any presentation, there are always at least as many relations s as generators r. The ideal generated by the $r \times r$ minors of the presentation matrix is independent of choices, and called the *fitting ideal*. Alexander [1] showed that for any knot K there is always a square presentation matrix, so that the fitting ideal is principal, and a generator for this ideal (normalized by multiplication by a suitable unit $t^{\pm n}$ so that it has no terms with negative exponent, and has a nonzero constant term) is the Alexander polynomial $\Delta_K(t)$.

5. Skein theory

5.1. **Kauffman bracket.** Up till now we have made extensive use of a diagrammatic calculus in which we perform local moves on a diagram (Reidemeister moves) which do not change the underlying knot type. Skein theory studies the *universal relations* that hold between *different* knots or links which are themselves related by a *different* set of local moves on a diagram.

Perhaps the simplest example of a move of the kind we mean is *resolving a crossing*. There are two distinct ways to do this, indicated in the figure.



As with Reidemeister moves, a resolution is meant to indicate the effect of a change to a diagram supported in a small region; the part of the diagram not drawn in the figure is supposed to be unchanged before and after the operation.

If D is a knot or link diagram, and if D_+ and D_- are obtained by making the first and second crossing resolutions indicated as above, then we can ask what is the relationship between D, D_+ and D_-. A *skein relation* is a universal relationship that holds — whatever D is and whatever crossing of D we resolved — between (the values of some invariant evaluated on) D, D_+ and D_-.

One particularly beautiful and important example is the *Kauffman bracket*. It is an invariant of a diagram taking values in the set of Laurent polynomials $\mathbb{Z}[A, A^{-1}]$ in a variable that by convention is usually denoted A. In other words, to any (topological equivalence class of) diagram D it associates a Laurent polynomial denoted $\langle D\rangle \in \mathbb{Z}[A, A^{-1}]$.

The Kauffman bracket satisfies the following 3 axioms:

(1) for any diagram D and for any other two diagrams D_+, D_- related by resolving any single crossing of D as above, we have the relation $\langle D\rangle = A\langle D_+\rangle + A^{-1}\langle D_-\rangle$. Or in more pictorial language:



(2) if D is any diagram, and D' is obtained from D by adding any single disjoint unknotted circle to the diagram, then $\langle D'\rangle = (-A^2 - A^{-2})\langle D\rangle$; and
(3) if D is the unknot then $\langle D\rangle = 1$.

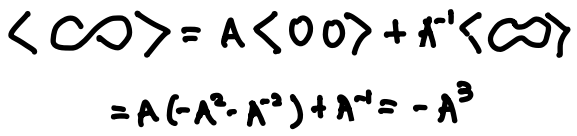
Exercise 5.1. Show that the Kauffman bracket is well-defined! In other words: show that it can be computed recursively by a finite application of the axioms, and that any two ways of computing it will give the same answer.

Let's compute some examples.

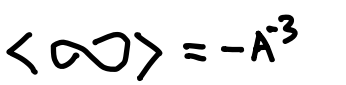
Example 5.2 (unlinks). Let D be any diagram that consists of n disjoint unknotted circles (i.e. there are no crossings at all). Then $\langle D\rangle = (-A^2 - A^{-2})^{n-1}$.

Notice that for fixed n all these diagrams represent the same link (the unlink on n components) but there are many inequivalent diagrams (essentially: the number of rooted trees with $n + 1$ vertices).

Example 5.3 (Reidemeister 1). Consider the following calculation:



The Kauffman bracket is not invariant under Reidemeister 1! Likewise:



Exercise 5.4. If D is any diagram, and D' is obtained from D by doing a positive (resp. negative) Reidemeister 1 move, then $\langle D' \rangle = -A^3 \langle D \rangle$ (resp. $\langle D' \rangle = -A^{-3} \langle D \rangle$).

Example 5.5 (Left handed trefoil). Let's compute the bracket for one of the standard diagrams for the left-handed trefoil. Resolving the top crossing gives two diagrams, one of an unknot with two positive Reidemeister 1 moves applied to it, and one of a Hopf link.



Resolving the left crossing for the Hopf link gives two unknots, one with a positive Reidemeister 1 move applied, and one with a negative Reidemeister 1 move applied.

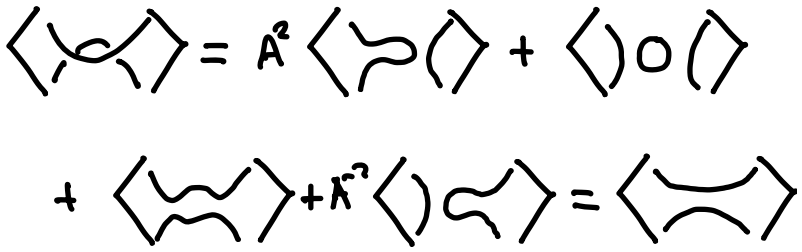


Putting this together, we see that

$$\langle \text{LH trefoil diagram} \rangle = A^7 + A^{-1}(-A^4 - A^{-4}) = A^7 - A^3 - A^{-5}$$

Proposition 5.6. *The Kauffman bracket is invariant under Reidemeister 2 and Reidemeister 3.*

Proof. This is an easy calculation. First we resolve the 2 crossings in a bigon in all four possible ways to obtain four terms. Three of these cancel, leaving the fourth term and proving invariance under Reidemeister 2.

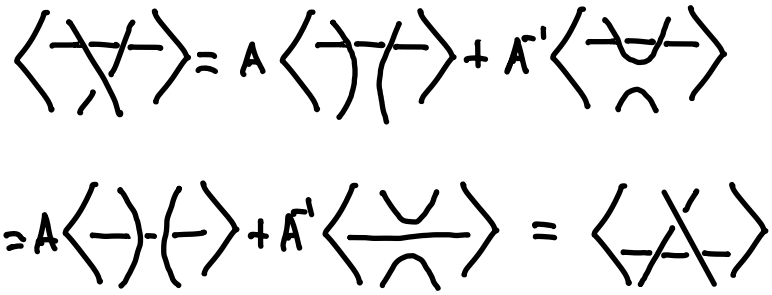


To prove invariance under Reidemeister 3, resolve the uppermost overcrossing (in one of the figures). Then use invariance under Reidemeister 2 to symmetrize the figures. 'Unresolving' the crossings in the two figures gives the result.

□

Thus: the Kauffman bracket is invariant under Reidemeister 2 and Reidemeister 3, and under Reidemeister 1 it is multiplied by $-A^{\pm 3}$. In order to arrive at a link invariant one may 'correct' the Kauffman bracket using another invariant of diagrams known as the *writhe*.

Recall that we defined the linking number of two oriented links by counting overcrossings of one component over the other, with sign. If we try to define a 'self-linking' number of a



single oriented diagram D of a knot by counting all overcrossings (with sign) we obtain an (integer-valued) invariant of a diagram called the *writhe*, and denoted $w(D)$. Changing the orientations of both components of an oriented 2-component link does not change linking number; likewise, changing the orientation of an oriented diagram of a knot does not change the writhe.

The writhe is invariant under Reidemeister 2 and 3, just as linking number is, and with the same proof. However, it is *not* invariant under Reidemeister 1: doing a positive Reidemeister 1 move increases the writhe by 1, and doing a negative Reidemeister 1 move decreases it by 1.

Corollary 5.7. *Let D be a knot diagram. Then $f_D(A) := (-A^3)^{-w(D)}\langle D\rangle$ is invariant under all three Reidemeister moves, and is therefore a knot invariant.*

Example 5.8 (Trefoils). Taking the mirror image of a knot interchanges the roles of A and A^{-1}. The writhe of the left handed trefoil is -3, so its invariant is $-A^{16}+A^{12}+A^4$ whereas the right handed trefoil has invariant $-A^{-16}+A^{-12}+A^{-4}$. These are different! We have finally managed to show that the left and right handed trefoils are different knots!

5.2. **Alternating knots and links.** The Kauffman bracket is not an invariant, but the *span* of the bracket — the difference between the highest and lowest powers of A that appear — *is* an invariant. In fact, the span of the bracket is an invariant of an unoriented *link*, even though the writhe of an unoriented link is *not* well-defined.

We shall shortly establish an inequality relating span and crossing number; what is remarkable about this inequality is that it is *sharp* for the class of so-called *alternating* knots and links.

But first, let's express the Kauffman bracket in a slightly different way.

Definition 5.9. For D a knot or link diagram, a *state* s of D is a function from the crossings of D to $\{-1, 1\}$.

In particular, for a diagram with c crossings there are 2^c different states. If D is a diagram and s is a state, we denote by $s(D)$ the diagram obtained from D by performing a resolution at each crossing p of D according to the sign of s: a positive resolution if $s(p) = 1$ and a negative resolution if $s(p) = -1$. Thus, $s(D)$ is a disjoint union of some number of unknotted components; we denote this number by $|s(D)|$. There is also a function from states to integers: if s is a state, we let $h(s) := \sum_p s(p)$ where the sum is taken over the crossings of D.

Lemma 5.10. *If D is a diagram, the Kauffman bracket is given by the formula*

$$\langle D \rangle = \sum_s A^{h(s)}(-A^2 - A^{-2})^{|s(D)|-1}$$

where the sum is taken over all states s of D.

The proof is immediate from the inductive definition of the bracket (compare with exercise 5.1).

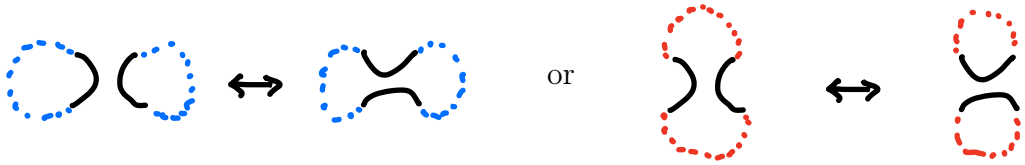
If D is any diagram, let s^+ be the state which sends every crossing to 1, and let s^- be the state which sends every crossing to -1.

Proposition 5.11. *Let D be a diagram. Then*

1. *if s and s' are states which differ at exactly one crossing, then $|s(D)|$ and $|s'(D)|$ differ by exactly 1; and*
2. *if H and L are the highest and lowest powers of A appearing in $\langle D \rangle$, then there are inequalities*

$$H \leq c + 2(|s^+(D)| - 1) \text{ and } L \geq -c - 2(|s^-(D)| - 1)$$

Proof. Let s and s' be two states that differ at exactly one crossing. There are two combinatorial possibilities for how the components of $s(D)$ and $s'(D)$ that meet at the crossing are connected to two pairs of edges (a third possibility is excluded because it would force the components to intersect). These possibilities are illustrated in the figure from which one can see that changing the crossing in either case changes the number of components by 1. This proves the first claim.



Lemma 5.10 gives a formula for the bracket as a sum of terms, one for each state of D. For any state s, the biggest power of A that appears in the term associated to s is $h(s) + 2(|s(D)| - 1)$. If s' is obtained from s' by changing the value at one crossing from 1 to -1 then $h(s') = h(s) - 2$ and $|s'(D)| \leq |s(D)| + 1$. It follows that the biggest power that appears in *any* term appears in the term associated to the state s^+, and the smallest power that appears in any term appears in the term associated to the state s^-. This proves the second claim. □

By Proposition 5.11 we obtain for any diagram an inequality $\text{span} = H - L \leq 2c + 2(|s^+(D)| + |s^-(D)| - 2)$. Our next task is to bound $|s^+(D)| + |s^-(D)|$ in terms of the crossing number. In full generality this is impossible, since (for example) a standard diagram of an unlink on n components has span $2n - 2$ and no crossings at all! We must therefore restrict attention to diagrams with a particular property.

Definition 5.12. A diagram is said to be *connected* if the underlying projection (consisting of an immersed circle for each component) is a connected subset of the plane.

Any diagram can be converted into a connected diagram by applying a series of Reidemeister 2 moves, which does not affect the bracket but increases the crossing number by a multiple of 2.

Definition 5.13. A link is said to be *split* if there are disjoint 3-balls so that we can put some (but not all) components in one of the 3-balls and the rest in the other.

Exercise 5.14. Show that a link is split if and only if it admits a diagram which is not connected.

Evidently, every diagram for a knot is connected.

Proposition 5.15. *For any connected diagram D there is an inequality $|s^+(D)|+|s^-(D)| \leq c+2$.*

Proof. Let's imagine the diagrams $s^-(D)$ and $s^+(D)$ lying on two parallel planes in $\mathbb{R}^3$, the diagram for $s^+(D)$ on a slightly higher plane. We may interpolate between these two diagrams by inserting a little saddle at each crossing of D that runs between the negative and the positive resolution, together with product rectangles that join up the rest of the diagrams. This builds a compact oriented *connected* surface Σ with $|s^+(D)| + |s^-(D)|$ boundary components, and therefore $\chi(\Sigma) \leq 2 - (|s^+(D)| + |s^-(D)|)$. On the other hand, each saddle contributes -1 to $\chi(\Sigma)$ and therefore $\chi(\Sigma) = -c$. The proposition follows. □

Corollary 5.16. *Suppose that L is a non-split link. Then the span of the Kauffman bracket of (any diagram for) L satisfies span $\leq 4c(L)$.*

Proof. For any diagram D representing L we have $H - L \leq 2c + 2(|s^+(D)| + |s^-(D)| - 2)$ by Proposition 5.11. On the other hand $|s^+(D)| + |s^-(D)| \leq c + 2$ by Proposition 5.15. It follows that $\text{span}\langle D\rangle = H - L \leq 4c$. □

When is equality achieved? By examining the arguments above it is clear that a sufficient condition for equality span $= 4c$ is that the following two conditions hold:

(1) for each state s obtained from s^+ (resp. s^-) by changing one crossing, $|s(D)| = |s^+(D)| - 1$ (resp. $|s(D)| = |s^-(D)| - 1$); and
(2) there is equality $|s^+(D)| + |s^-(D)| = c + 2$.

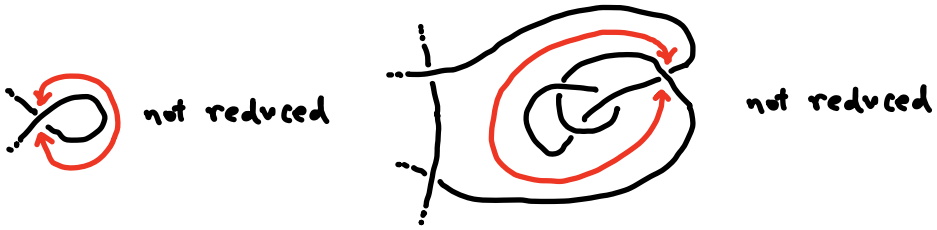
We shall show that both these conditions hold for the class of alternating links.

Definition 5.17. A diagram D is *alternating* if, as we move along each component in any direction, the crossings alternate between overcrossings and undercrossings. A diagram is *reduced* if there is no complementary region which meets the same crossing more than once.

In other words, the diagram is alternating if successive crossings on every component look like this:



and the diagram is reduced if it doesn't have two connected subsets joined to each other at a single crossing like this:



Any diagram for a knot or link with minimal crossing number must be reduced. For, if a diagram D is not reduced, so that there is a crossing p separating pieces D_1, D_2 of D, we may take one piece (e.g. D_2) and 'flip it over' at p to produce a new diagram with one fewer crossing.

We are now in a position to define the class of alternating links:

Definition 5.18. A link is *alternating* if it has a diagram which is connected, reduced and alternating.

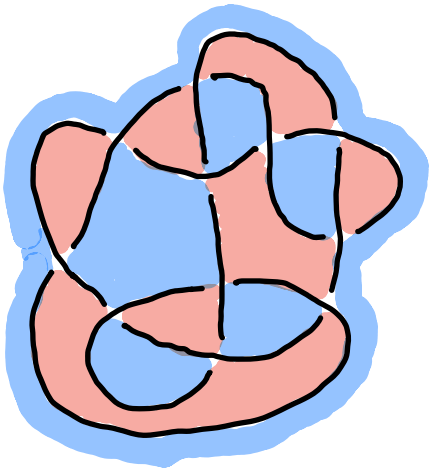
The trefoil and figure 8 knots are alternating knots. In fact, every knot with fewer than 8 crossings is alternating! It is a theorem of Menasco that an alternating link is non-split; in particular, *every* diagram for an alternating link is connected (of course, this is immediate for alternating *knots* since no knot can ever be split).

We are now in a position to prove the main theorem we have been building up to:

Theorem 5.19. *Let L be an alternating link, and let D be any connected, reduced and alternating diagram for L. Then span $= 4c(D)$ and consequently span $= 4c(L)$.*

Proof. Let D be any connected, reduced and alternating diagram for L. We have already shown span $\leq 4c(D)$. As pointed out earlier, to prove equality it suffices to establish the following two properties:

(1) for each state s obtained from s^+ (resp. s^-) by changing one crossing, $|s(D)| = |s^+(D)| - 1$ (resp. $|s(D)| = |s^-(D)| - 1$); and
(2) there is equality $|s^+(D)| + |s^-(D)| = c + 2$.



Since D is connected and alternating, we may color the complementary components red and blue so that around the boundary of each red region the overcrossings go anticlockwise, while around the boundary of each blue region the overcrossings go clockwise.

Once we have done this, it is easy to see that when we resolve all the crossings positively the components of $s^+(D)$ become the boundaries of the red regions, whereas when we resolve all the crossings negatively the components of $s^-(D)$ become the boundaries of the blue regions. The fact that the diagram is connected implies that each complementary region is a disk (except the outside region, which is an annulus) so that the total number of complementary regions is equal to $c+2$ and we have proved the second property.

Furthermore, if s is obtained from s^+ by changing one crossing, at this crossing two distinct red regions are catecornered (the fact that the regions are *distinct* follows from the fact that the diagram is *reduced*!) and therefore changing this crossing connects up the two boundary circles into a single circle of $s(D)$. Hence $|s(D)| = |s^+(D)| - 1$ (and we may argue likewise for s^-) and we have proved the first property.

It follows as in the proof of Proposition 5.11 that the highest power of A that appears in any term in the formula for the bracket appears *only* in the term associated to $s^+(D)$, and the lowest power appears *only* in the term associated to $s^-(D)$. Thus the inequalities in Proposition 5.11 are equalities. Since we have just shown $|s^+(D)| + |s^-(D)| = c+2$ the theorem is proved. □

This theorem was proved by Louis Kauffman in the paper [6] in which he introduced his bracket, and independently (using similar combinatorial ideas) by Kunio Murasugi and Morwen Thistlethwaite.

A remarkable corollary of this theorem is that for an alternating knot or link, *any* reduced alternating diagram has the minimal crossing number (amongst *all* diagrams) and in particular, any two such diagrams have the *same* number of crossings! Kauffman's bracket is powerful indeed!

5.3. **HOMFLY.** Suppose D is an oriented link diagram, and p is a crossing. Only one resolution of the crossing is compatible with the orientation; this gives rise to a new diagram D^0 with one fewer crossing. However, there are *two* oriented diagrams that give rise to D^0 by resolving one crossing: D, and the diagram obtained from D by changing the overcrossing at p into an undercrossing. Thus we are led to consider triples of oriented diagrams D^+, D^-, D^0 that are related by the following local modification:



We may (try to!) define a (Laurent) polynomial invariant P of oriented diagrams that is invariant under Reidemeister moves (and therefore defines an invariant of oriented *links*) and that satisfies a *skein relation* of the form

$$xP(L^+) + yP(L^-) + zP(L^0)$$

for suitable variables x, y, z and where L^+, L^-, L^0 are the oriented links represented by any three diagrams of the form D^+, D^-, D^0 as above. Since the skein relation is linear in P we may normalize the invariant P (if one exists) by setting $P(\text{unknot}) = 1$. Furthermore, since it is homogeneous in x, y, z we may eliminate one of the variables and get an equivalent skein relation. One common normalization is:

$$\alpha P(L^+) - \alpha^{-1} P(L^-) = z P(L^0)$$

so that we are looking for a two-variable Laurent polynomial invariant P in α and z.

Leaving aside the question of existence for the moment, we can at least prove uniqueness:

Proposition 5.20. *There is at most one polynomial invariant P of oriented links with $P(\text{unknot}) = 1$ that satisfies the skein relation $\alpha P(L^+) - \alpha^{-1} P(L^-) = z P(L^0)$.*

Proof. The following figure shows that $P = z^{-1}(\alpha - \alpha^{-1})$ for the unlink on two components. A similar computation lets us determine the value of P on the unlink with any number of components.



Now if L is an oriented link represented by a diagram D we claim that there is another diagram D' obtained from D by changing some number of crossings, that represents the unlink. First of all, we may change the crossings so that one component of L is 'underneath' all the others; thus the claim will follow by induction for links if we can prove it for knots.

A knot diagram D is obtained from an immersed circle γ in the plane by making a choice of crossing at each self-intersection of γ. Given γ, pick a point p on γ which is not a self-intersection, and then let $\hat{\gamma}$ be a path in $\mathbb{R}^3$ whose z coordinate is steadily increasing, and whose projection starts and ends at p and traces once over γ. We may close up $\hat{\gamma}$ to a knot K by joining the two points that project to p by a vertical arc δ, and we may then perturb $\hat{\gamma}$ slightly so that its projection is very close to γ; in particular, it is represented by a diagram D' obtained from D by changing some of the crossings.

We claim K is an unknot. To see this, observe that K is the union of two arcs $\hat{\gamma}$ and δ, on each of which the z coordinate is steadily increasing (at least if we orient the arcs appropriately). We may therefore join points on δ and $\hat{\gamma}$ with the same z coordinate by a straight horizontal line segment, and then observe that the union of these segments sweeps out an embedded disk in $\mathbb{R}^3$ with boundary K. Thus K bounds a disk, and is therefore the unknot. The claim is proved.

It follows that for each oriented link type L we may associate a *complexity*, which consists of a pair of non-negative integers (c, n) where c is the crossing number of L, and where n is the minimum, over all diagrams for L with crossing number c, of the number of crossing changes needed to change L into an unlink. We order the set of possible values of the complexity lexicographically: (c_1, n_1) is less than (c_2, n_2) if and only if either $c_1 < c_2$, or $c_1 = c_2$ and $n_1 < n_2$. The set of values of the complexity is evidently well-ordered with (unique) minimum $(0, 0)$ represented only by unlinks.

Now, if L is any link, and if D is a diagram for L realizing the minimal complexity, we may change some crossing of D to obtain a new diagram D' of strictly lower complexity.

If $D+, D^-, D^0$ are the three diagrams associated to this crossing, then the skein relation computes $P(L)$ in terms of the values of P on links of strictly lower complexity. Since P is determined on unlinks, this proves that P is determined on L, and we are done. □

If one is careful, one can associate in a systematic way to each oriented diagram D, a specific sequence of crossing changes that assign (inductively) a specific value $P(D)$ to D via the skein relation. Then one can try to prove that this polynomial invariant of oriented diagrams is invariant under Reidemeister moves. Rather remarkably, this procedure works, so that there really is a (2-variable) polynomial invariant P of oriented links satisfying the skein relation. It was discovered independently and at virtually the same time by six people (four working in two groups of two), and is therefore named the HOMFLY polynomial after the initials of their surnames.

Two specializations of this polynomial are particularly interesting.

Example 5.21 (Alexander polynomial). Taking $\alpha = 1$ and $z = t^{1/2} - t^{-1/2}$ gives rise to the Alexander polynomial! Thus we may compute the Alexander polynomial via the skein relation

$$\Delta_{L^+}(t) - \Delta_{L^-}(t) = (t^{1/2} - t^{-1/2})\Delta_{L^0}(t)$$

Since performing a resolution of a crossing changes the parity of the number of components by 1, only integer powers of t appear for knots. Notice too that this gives a polynomial invariant for oriented links, which may be computed in terms of t-colorings, just like the Alexander polynomial.

This skein relation definition of the Alexander polynomial has a curious history. It was first discovered by Alexander, and appeared in his paper [1] towards the end, in a section titled 'miscellaneous theorems', but it seems to have been largely forgotten. It was rediscovered by Conway [2] in the 1960s, who introduced the term 'skein relation' (Conway's version uses z in place of $t^{1/2} - t^{-1/2}$; with this normalization one talks about the 'Conway polynomial'.)

Example 5.22 (Jones polynomial). Taking $\alpha = t^{-1}$ and $z = t^{1/2} - t^{-1/2}$ gives rise to the *Jones polynomial* V, which may (therefore) be computed via the skein relation

$$t^{-1}V_{L^+}(t) - tV_{L^-}(t) = (t^{1/2} - t^{-1/2})V_{L^0}(t)$$

The Jones polynomial is equivalent to the polynomial invariant $f_D(A) = (-A^3)^{-w(D)}\langle D\rangle$ of an oriented link via the substitution $A = t^{-1/4}$.

Exercise 5.23. Prove that the invariant $V_L(t)$ obtained from $f_D(A) = (-A^3)^{-w(D)}\langle D\rangle$ via the substitution $A = t^{-1/4}$ does indeed satisfy the skein relation

$$t^{-1}V_{L^+}(t) - tV_{L^-}(t) = (t^{1/2} - t^{-1/2})V_{L^0}(t)$$

In fact, the exposition in these notes gets the history round the wrong way. Jones discovered his invariant in 1984 out of his work in the theory of operator algebras [5]. The fact that this work gave rise to an invariant of knots came as a complete surprise to Jones and to everyone else; and its discovery led to a revolution in knot theory. Kauffman's definition of the bracket emerged soon after as a way to clarify the meaning of the Jones polynomial (and to give a simple proof of its invariance), and once people understood

that skein relations could give rise to more than one polynomial invariant, the HOMFLY polynomial was discovered very quickly.

In the 40 years since the discovery of the Jones polynomial, knot theory has developed into a remarkably deep and sophisticated subject, with significant applications to theoretical physics, the biology of DNA, and many other fields.

Mathematically speaking, the theory of irreducible representations of the knot group into $\mathrm{SL}(2,\mathbb{C})$ turns out to be intimately connected to hyperbolic geometry, and the geometrization conjecture of Thurston (proved as it applies to knots and links by Thurston, and in full generality by Perelman).

Polynomial invariants such as HOMFLY are subsumed under the auspices of Khovanov Homology (which sheds light on exotic smooth structures on 4-manifolds) and Heegaard Floer Homology (which has its foundations in symplectic topology and Gromov's theory of pseudo-holomorphic curves, and which is by now connected to many different aspects of 3 and 4-manifold topology).

6. Acknowledgments

While producing these notes I consulted (with their generous permission) notes taken by Juliet Oliver and Parth Wokhlu during the course I taught at the University of Chicago Center in Paris from March 31 to April 17 2025. I have also borrowed liberally from the resources listed in the bibliography (especially [7]).

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